

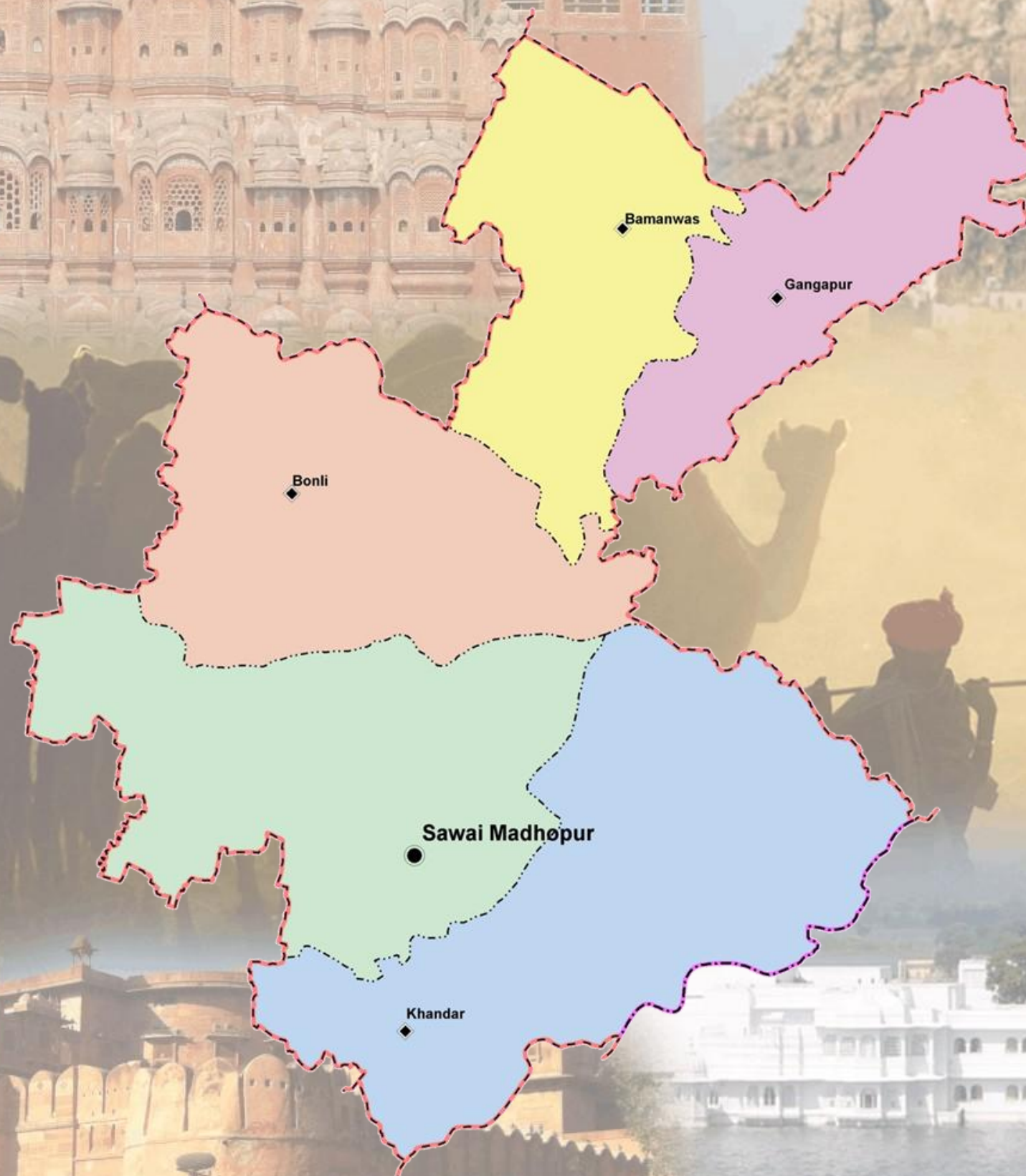


Ground Water Department,  
Rajasthan

# Hydrogeological Atlas of Rajasthan Sawai Madhopur District



European Union  
State Partnership Programme



2013



Rolta India Limited



## Message



**Shri Ashok Gehlot,**  
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

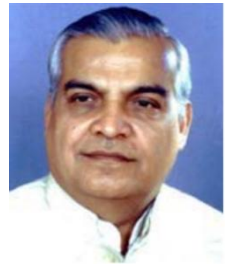
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of 'Hydrogeological Atlas of Rajasthan'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

**(Ashok Gehlot)**

## Message



**Dr. Jitendra Singh,**  
Hon'ble Minister for Energy,  
Non-Conventional Energy Resources,  
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

**(Jitendra Singh)**

## Message



**Dr. Purushottam Agarwal**

Principal Secretary to Government,  
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of 'Hydrogeological Atlas of Rajasthan' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

**(Purushottam Agarwal)**

## Foreword



**Arno Schaefer**

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

**(Arno Schaefer)**

# Hydrogeological Atlas of Rajasthan

## Sawai Madhopur District

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## ADMINISTRATIVE SETUP

## DISTRICT – SAWAI MADHOPUR

### Location:

Sawai Madhopur district is located in the eastern part of Rajasthan. It is bounded in the north by Dausa and northeast by Karauli districts, in the east by state of Madhya Pradesh, South by Kota while Tonk district constitutes its western boundary. It stretches between 25° 44' 00.90" to 26° 43' 34.33" north latitude and 75° 58' 36.70" to 76° 59' 04.98" east longitude covering area of 5,051.9 sq kms. The district is part of three river basins viz. 'Banas River Basin', 'Chambal River Basin' and 'Gambhir River Basin'. Of these, the Banas River basin is most prominent and drains significantly large part of the district in central and northern parts whereas the Chambal River drains in southern part of the district. The Gambhir river basin occupies small area in the northeastern corner.

### Administrative Set-up:

Sawai Madhopur district is administratively divided into five blocks. The following table summarizes the basic statistics of the district at block level.

| S. No.       | Block Name     | Population<br>(Based on 2001 census) | Area<br>(sq km) | % of District<br>Area | Total Number of<br>Towns and Villages |
|--------------|----------------|--------------------------------------|-----------------|-----------------------|---------------------------------------|
| 1            | Bamanwas       | 1,49,429                             | 772.6           | 15.3                  | 150                                   |
| 2            | Bonli          | 2,09,833                             | 1,049.0         | 20.7                  | 180                                   |
| 3            | Gangapur       | 2,84,605                             | 633.2           | 12.5                  | 124                                   |
| 4            | Khandar        | 1,55,383                             | 1,392.4         | 27.5                  | 181                                   |
| 5            | Sawai Madhopur | 3,13,303                             | 1,204.7         | 24                    | 163                                   |
| <b>Total</b> |                | <b>11,12,553</b>                     | <b>5,051.9</b>  | <b>100.0</b>          | <b>798</b>                            |

The district has 798 towns and villages of which five are block headquarters as well.

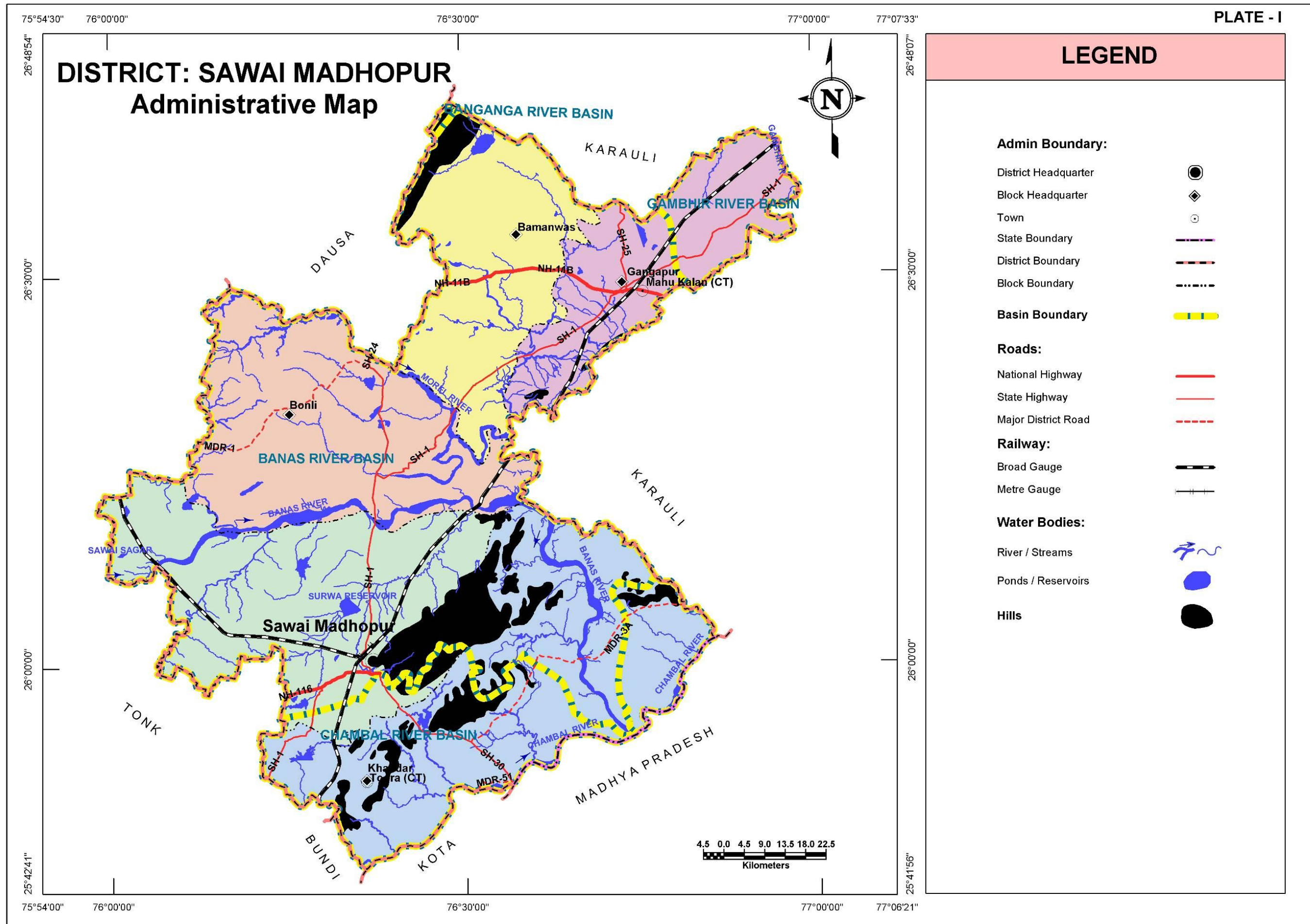
### Climate:

The climate of the district can be classified as sub-humid. It is characterized by very hot summers and very cold winters with fairly good rainfall during south-west monsoon period. In May, the maximum temperature may sometimes go up to 47 °C and winter temperatures dipping down to 2 °C. The potential evapotranspiration rates are high, especially during May and June. The mean annual rainfall of the district is 606.6mm. The most of the rainfall is received during the monsoon months.





PLATE - I



## TOPOGRAPHY

The district can be divided into three distinct physiographic units, viz. hilly terrain, alluvial plain with isolated hills and alluvial plains. The hilly terrain occupies the south and south eastern part with NE-SW trending ridges. The isolated hill occupies southwest and central part of the district and alluvial plain area occupies north, northeast and western parts of the district having relatively flat and gently sloping topography. The major rivers of the district are Banas, Chambal and Morel which creates very good drainage system along with their tributaries. The general topographic elevation in the district is between 250 m to 300 m above mean sea level. Elevation ranges from a minimum of 164.4 m above mean sea level in Khandar block in the southern part of the district to a maximum of 541.4 m above mean sea level in Bamanwas block in northern part of the district.

**Table: Block wise minimum and maximum elevation**

| S. No. | Block Name     | Min. Elevation<br>(m amsl) | Max. Elevation<br>(m amsl) |
|--------|----------------|----------------------------|----------------------------|
| 1      | Bamanwas       | 218.7                      | 541.4                      |
| 2      | Bonli          | 204.2                      | 449.3                      |
| 3      | Gangapur       | 219.1                      | 428.6                      |
| 4      | Khandar        | 164.4                      | 492.7                      |
| 5      | Sawai Madhopur | 213.0                      | 505.0                      |

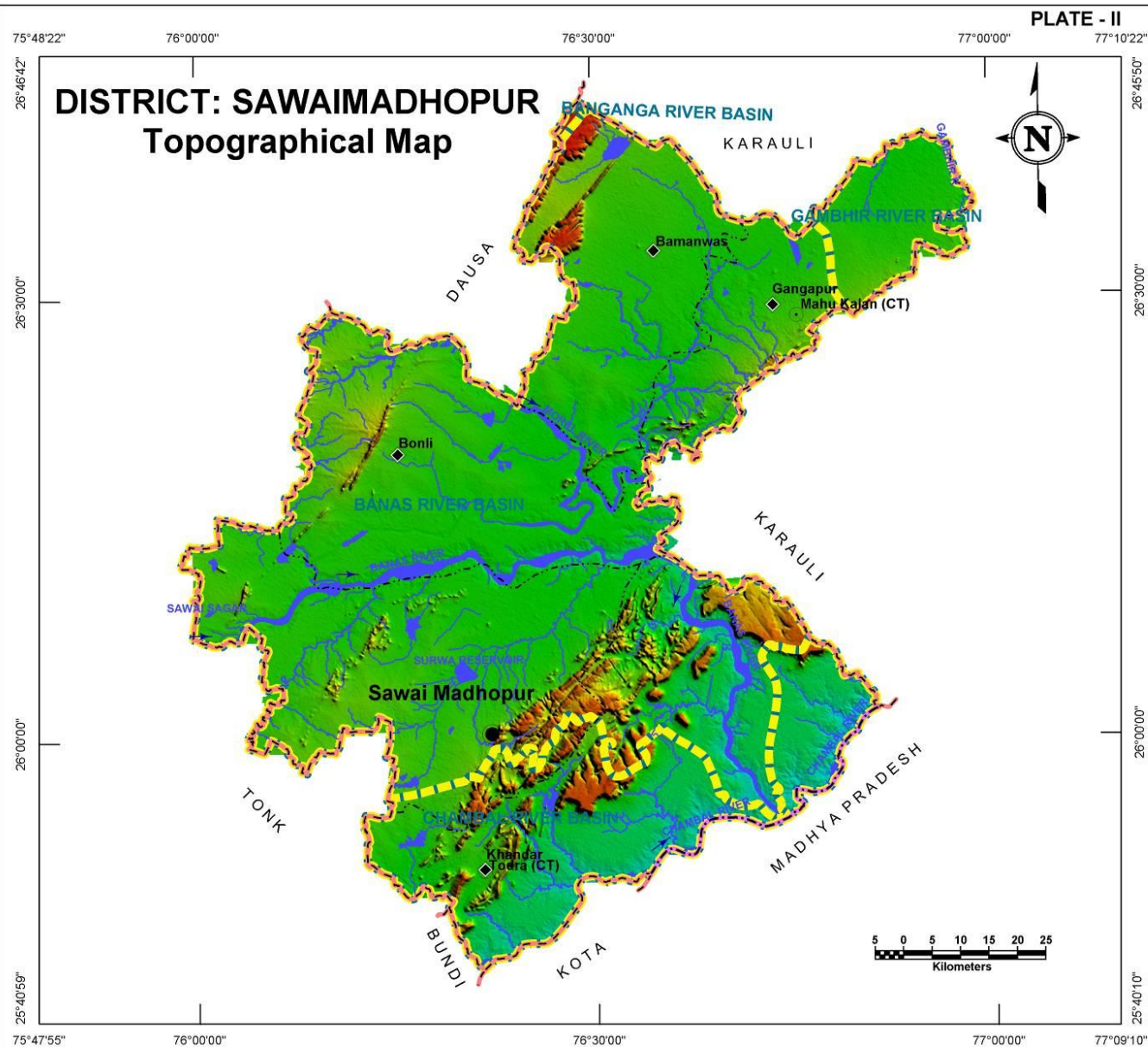
## RAINFALL

The district receives very good rainfall. The general distribution of rainfall across can be visualized from isohyets presented in the Plate – III. Low rainfall is seen in northeastern and western part of the district which gradually increases towards central and southern areas. The general variation of the rainfall in the district has been in the range of 700 to 1000 mm rainfall in the year 2010. The annual average rainfall was 838.6 mm based on the data of available blocks. Highest annual rainfall was in Bamanwas block (1,175.3 mm) whereas lowest in Bonli block (595.8 mm).

**Table: Block wise annual rainfall statistics (derived from year 2010 meteorological station data)**

| Block Name     | Minimum Annual<br>Rainfall (mm) | Maximum Annual<br>Rainfall (mm) | Average Annual<br>Rainfall (mm) |
|----------------|---------------------------------|---------------------------------|---------------------------------|
| Bamanwas       | 657.2                           | 1,175.3                         | 908.1                           |
| Bonli          | 595.8                           | 1,006.7                         | 793.8                           |
| Gangapur       | 619.5                           | 982.9                           | 755.6                           |
| Khandar        | 711.4                           | 1,060.5                         | 854.5                           |
| Sawai Madhopur | 663.5                           | 1,119.2                         | 880.9                           |





LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

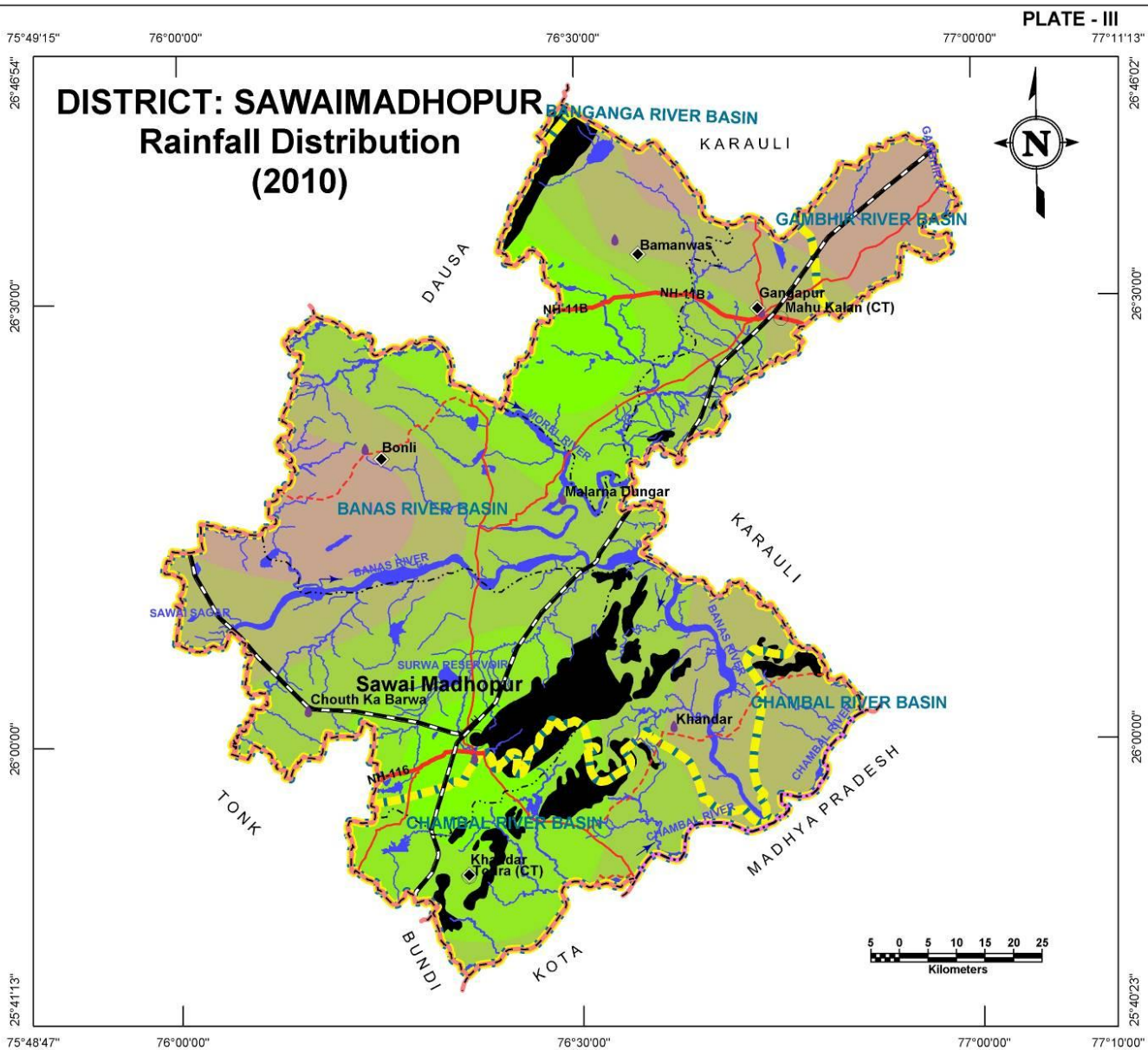
Water Bodies:

- River / Streams
- Ponds / Reservoirs

Elevation (m amsl):



Source : SRTM DEM



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- Raingauge Stations
- State Boundary
- District Boundary
- Block Boundary

Basin Boundary

Roads:

- National Highway
- State Highway
- Major District Road

Railway:

- Broad Gauge
- Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Isohyets (mm):

- 500-600
- 600-700
- 700-800
- 800-900
- 900-1000
- > 1000



## GEOLOGY AND GEOMORPHOLOGY

### Geology

The district is mostly covered with fluvial and wind-blown sand which as seen in north, northeast and central parts of the district. The district exposes rocks of Vindhyan, Delhi and Bhilwara Super Group. The Vindhyan Super Group is divided into Bhandar, Rewa, Sand and Satola Groups which consist of sandstone, shale and limestone. Alwar Group is a part of Delhi Super Group which occupies the northwestern part of the district. Alwar Group consists of quartzite, conglomerate and mica schist. Bhilwara Super Group is divided into Ranthambhore, Jahazpur, Hindoli and Mangalwar complex which is exposed in the west and southwestern parts of the district. Bhilwara Super Group consists of quartzite, slate, dolomite, phyllite, schist and amphibolite rock formation.

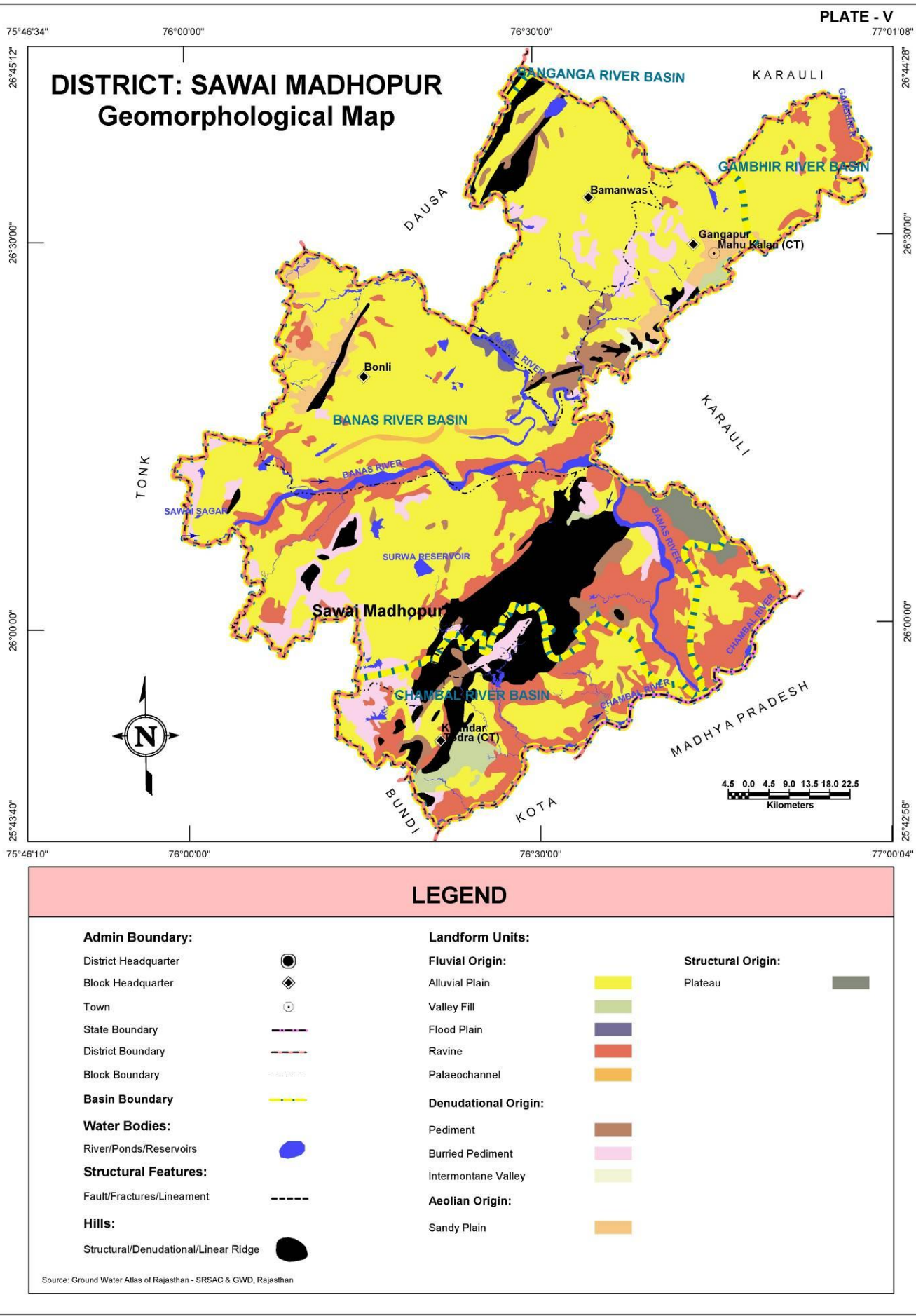
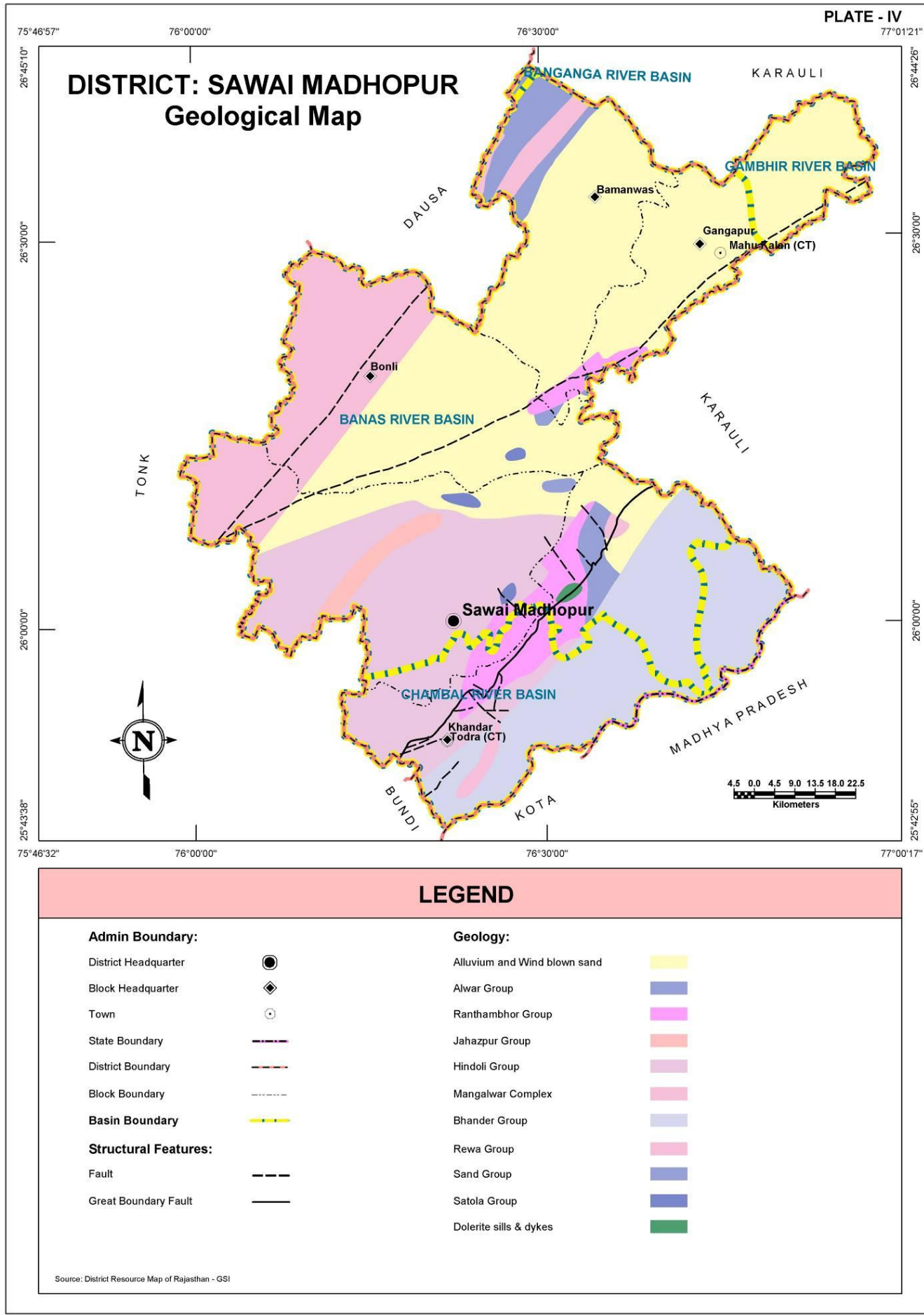
| Super Group | Group                   | Formation                                                                                                                                |
|-------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|             | Recent to Sub-Recent    | Dune sand, soil and Alluvium                                                                                                             |
|             | --X-----X-----X-----X-- | Unconformity--X-----X-----X-----X--                                                                                                      |
| Vindhyan    | Bhandar                 | Limestones and Shales                                                                                                                    |
|             | Rewa                    | Conglomerate                                                                                                                             |
|             | Sand                    | Limestone, Shale, Sandstone, Siltstone                                                                                                   |
|             | Satola                  |                                                                                                                                          |
|             | --X-----X-----X-----X-- | Unconformity--X-----X-----X-----X--                                                                                                      |
|             | Semri                   | Limestone, Conglomerate                                                                                                                  |
|             | --X-----X-----X-----X-- | Unconformity--X-----X-----X-----X--                                                                                                      |
| Delhi       | Alwar                   | Quartzite/Conglomerate and Mica Schist                                                                                                   |
|             | --X-----X-----X-----X-- | Unconformity--X-----X-----X-----X--                                                                                                      |
| Bhilwara    | Ranthambore             | Quartzite, Shales, Slates                                                                                                                |
|             |                         | Dolerite sills & dykes                                                                                                                   |
|             | Jahazpur                | Dolomite, ferruginous, chert, carbonaceous, phyllite, ferruginous phyllites with thin band of conglomerate, gritty quartzite & quartzite |
|             | Hindoli                 | Shale, Slate, Schist, Quartzite, Phyllite                                                                                                |
|             | Mangalwar Complex       | Migmatites gneiss, garnetiferous mica schist, sillimanite mica-schist, impure marble and amphibolite                                     |

## DISTRICT – SAWAI MADHOPUR

### Geomorphology

Table: Geomorphologic units, their description and distribution

| Origin       | Landform Unit                               | Description                                                                                                                                                                                                                                                                                                                                                       |
|--------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aeolian      | Sandy Plain                                 | Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.                                                                                                                                                                                                                         |
| Denudational | Buried Pediment                             | Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.                                                                                                                                                                                                                                                                   |
|              | Intermontane Valley                         | Depression between mountains, generally broad & linear, filled with colluvial deposits.                                                                                                                                                                                                                                                                           |
|              | Pediment                                    | Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.                                                                                                                                                                                                 |
| Fluvial      | Alluvial Plain                              | Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.                                                                                                                                                                                  |
|              | Flood Plain                                 | The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.                                                                                                                                                                          |
|              | Paleochannel                                | Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.                                                                                                                                                                                                                                                        |
|              | Valley Fill                                 | Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.                                                                                                                                                                          |
|              | Ravine                                      | Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.                                                                                                                                                                                                                                                        |
| Structural   | Plateau                                     | Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.                                                                                                                           |
| Hills        | Denudational, Structural Hill, Linear Ridge | Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike. |





## AQUIFERS

## DISTRICT – SAWAI MADHOPUR

Wide variety of lithologic material forms aquifers in the district (Plate – VI). Sandy and silty constituents within Older Alluvium form aquifers in unconsolidated sediments which occupy about 54% of district area and distributed as a central NE-SW trending belt running parallel and east of Aravali range as well as occurring as a large patch in the eastern part of the district along with some isolated clusters near western border. Weathered, fractured and jointed openings in hard rocks along with solution cavities specifically in limestone create secondary openings that store ground water. Among hard rock aquifers, Phyllites are most widespread, occupying about 17% of the district area and present in western part of the district. Other important hard rock aquifers are Shale, Limestone and Quartzite in order of spatial coverage and seen in central and eastern part of the district.

**Table: aquifer potential zones their area and their description**

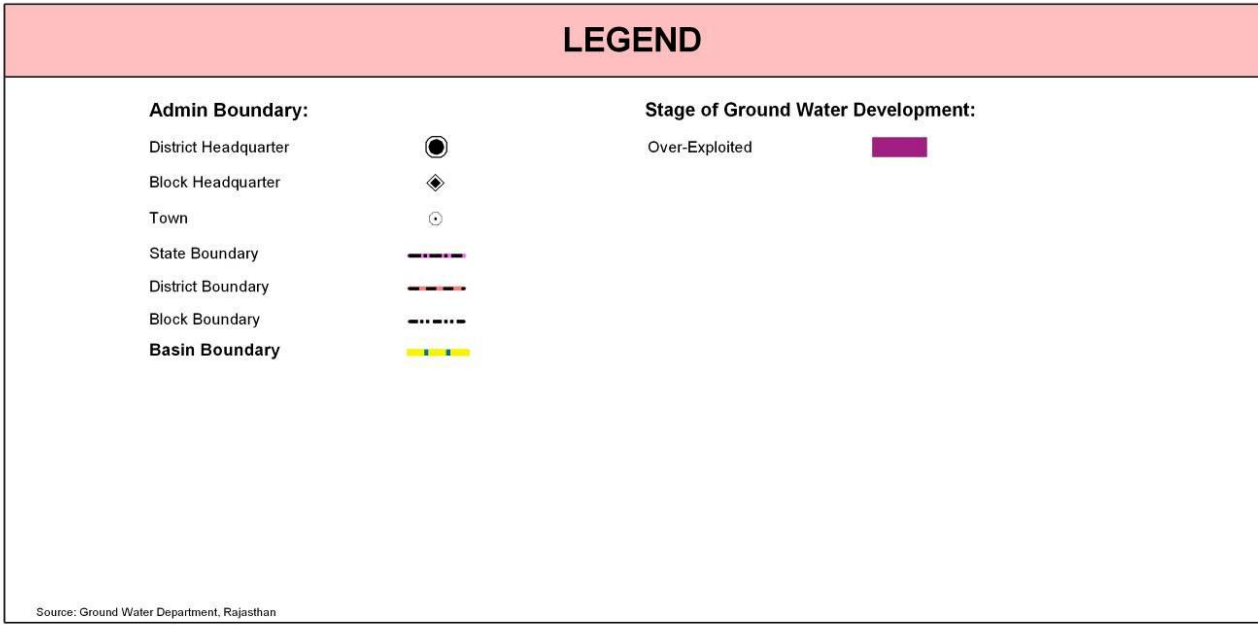
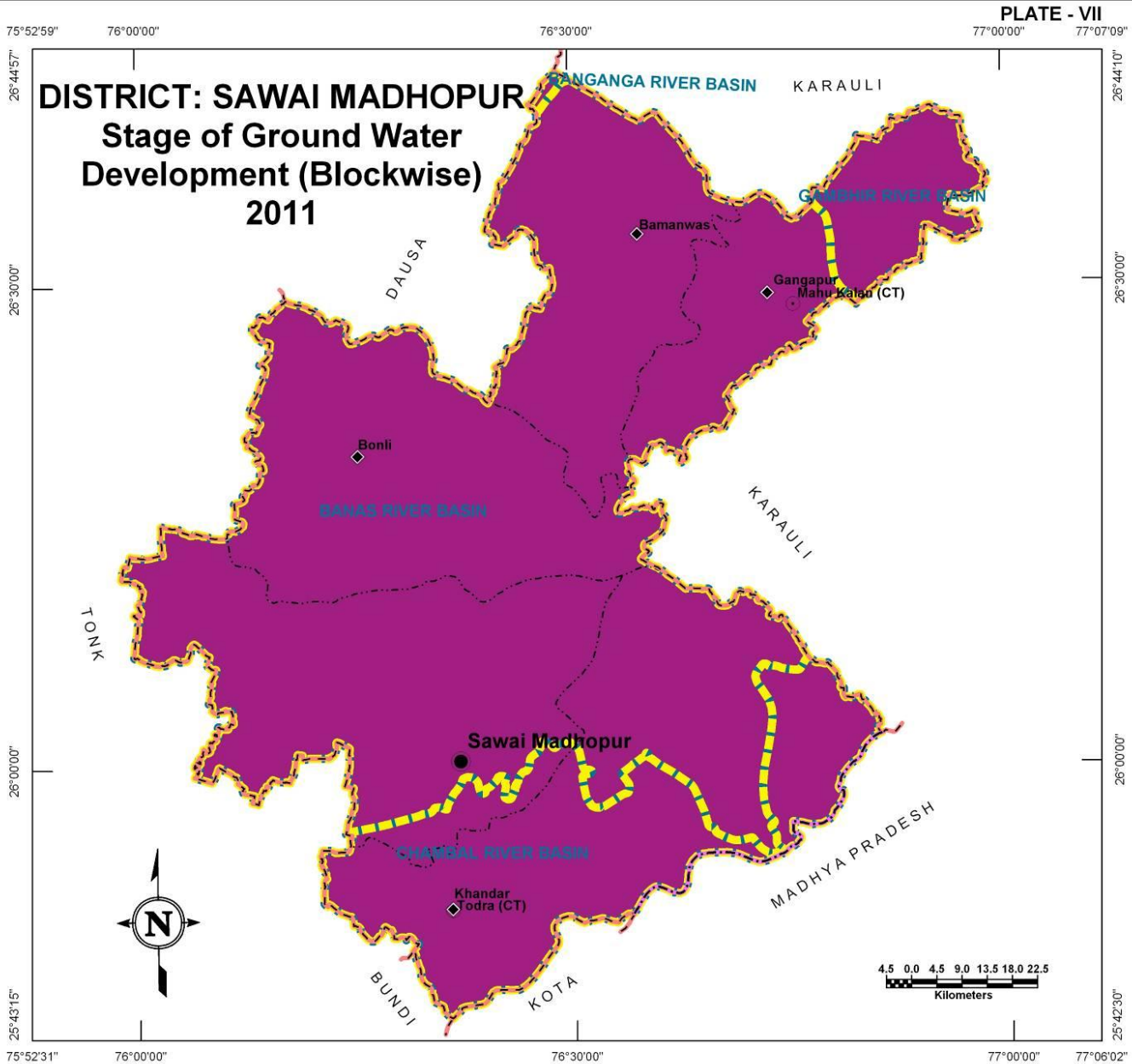
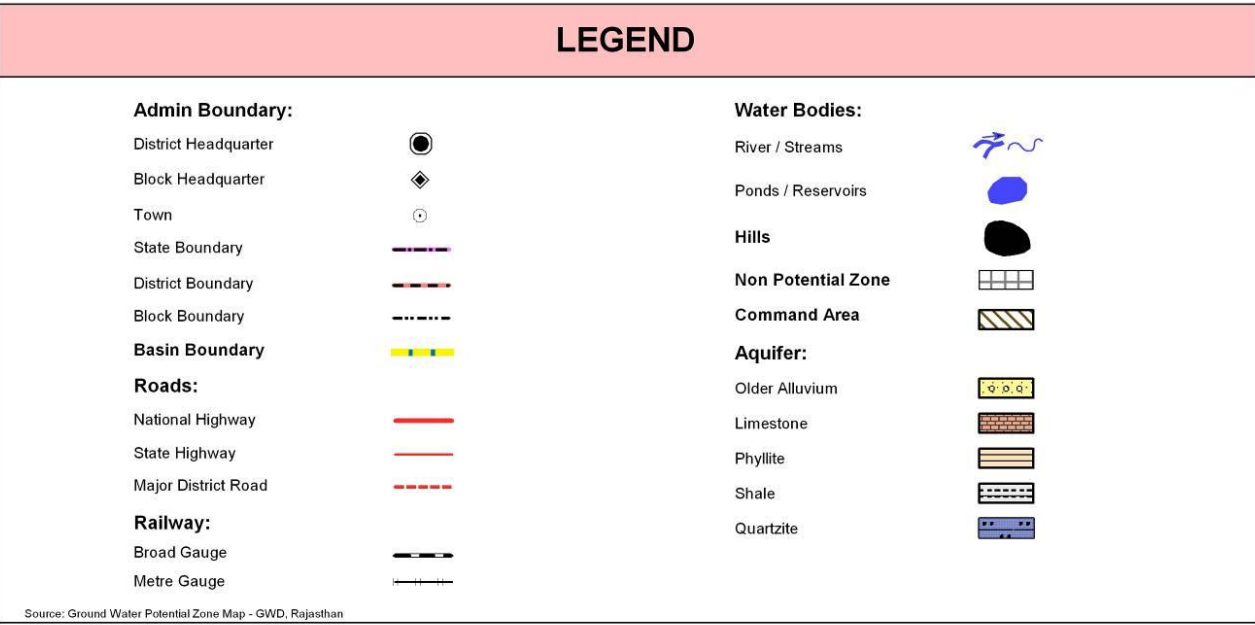
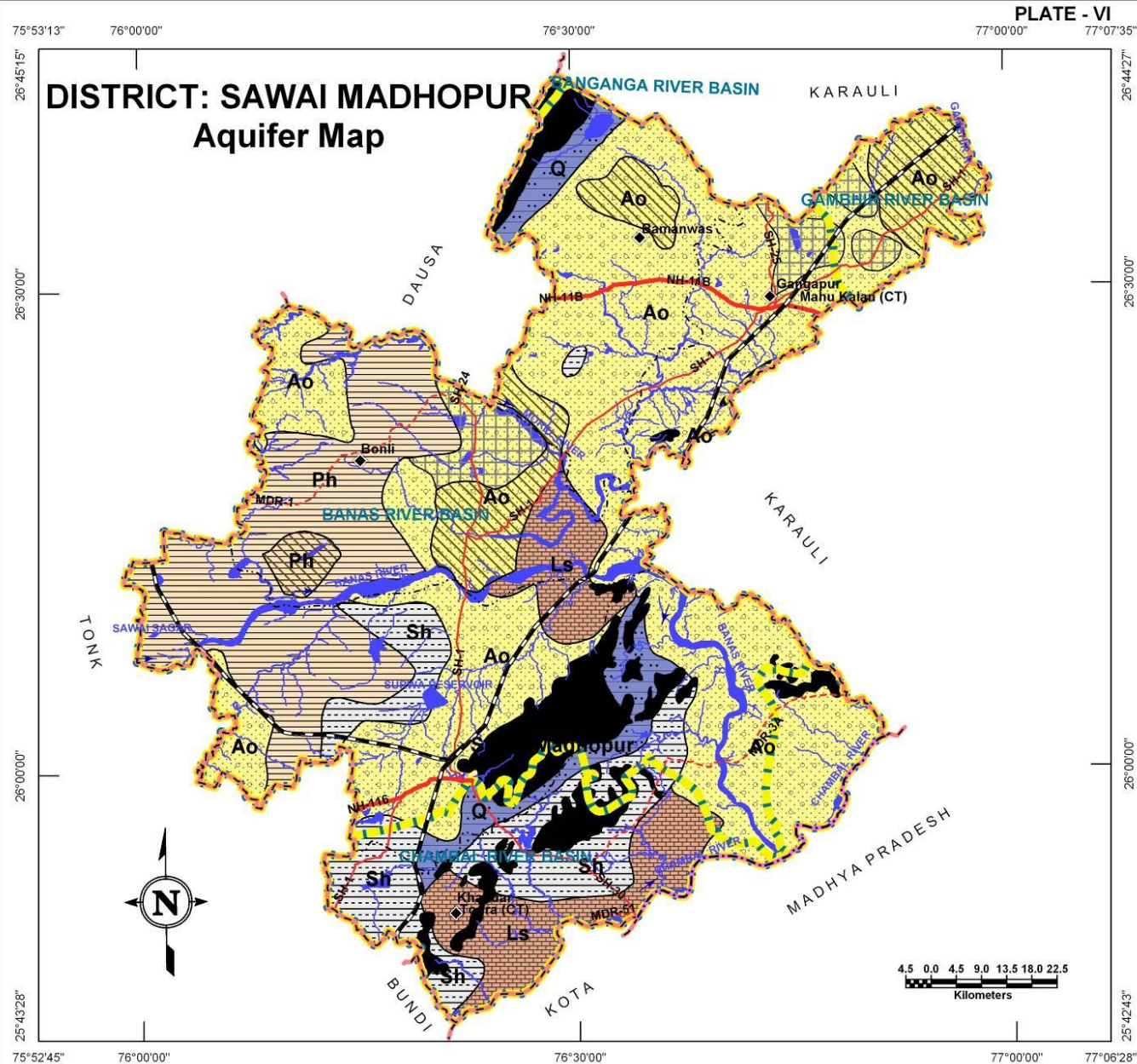
| Aquifer in Potential Zone | Area (sq km)   | % age of district | Description of the unit/Occurrence                                                                  |
|---------------------------|----------------|-------------------|-----------------------------------------------------------------------------------------------------|
| Older Alluvium            | 2,707.7        | 53.6              | This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar. |
| Limestone                 | 410.1          | 8.2               | In general, it is fine to medium grained, grey, red yellowish, pink or buff in colour.              |
| Phyllite                  | 876.2          | 17.3              | These include meta sediments and represented by carbonaceous phyllite.                              |
| Shale                     | 497.1          | 9.8               | Grey, light green and purple in colour and mostly splintery in nature.                              |
| Quartzite                 | 181.1          | 3.6               | Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.                   |
| Hills                     | 379.7          | 7.5               |                                                                                                     |
| <b>Total</b>              | <b>5,051.9</b> | <b>100.0</b>      |                                                                                                     |

## STAGE OF GROUND WATER DEVELOPMENT

Being a predominantly fresh ground water area coupled with good population and availability of agriculture land, the ground water exploitation is also very high and the ground water resource assessment studies reveal that all the blocks of the district have already exhausted the dynamic ground water resource and all the blocks fall within 'Over Exploited' category on the basis of stage of development exceeding more than 100%.

| Categorization on the basis of stage of development of GW | Block Name                                         |
|-----------------------------------------------------------|----------------------------------------------------|
| Over Exploited                                            | Bamanwas, Bonli, Gangapur, Sawai Madhopur, Khandar |

**Basis for categorization:** Ground water development >100% - Over-Exploited.





## LOCATION OF EXPLORATORY AND GROUND WATER MONITORING WELLS

## DISTRICT – SAWAI MADHOPUR

Sawai Madhopur district has a well distributed network of exploratory wells (106) and ground water monitoring stations (90) in the district owned by RGWD (56 and 66 respectively) and CGWB (50 and 24 respectively). The exploratory wells have formed the basis for delineation of subsurface aquifer distribution scenario in three dimensions. Benchmarking and optimization studies suggest that for optimal ground water level monitoring just one additional well is required to be added in Khandar block and for water quality monitoring, 115 additional wells must be added to the network for strengthening.

**Table: Block wise count of wells (existing and recommended)**

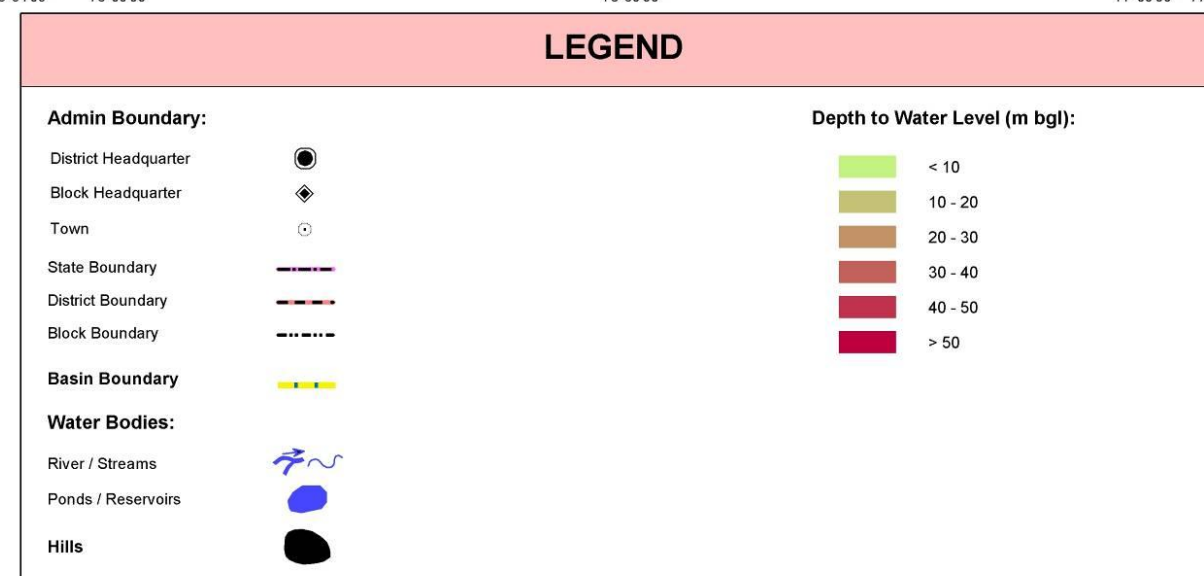
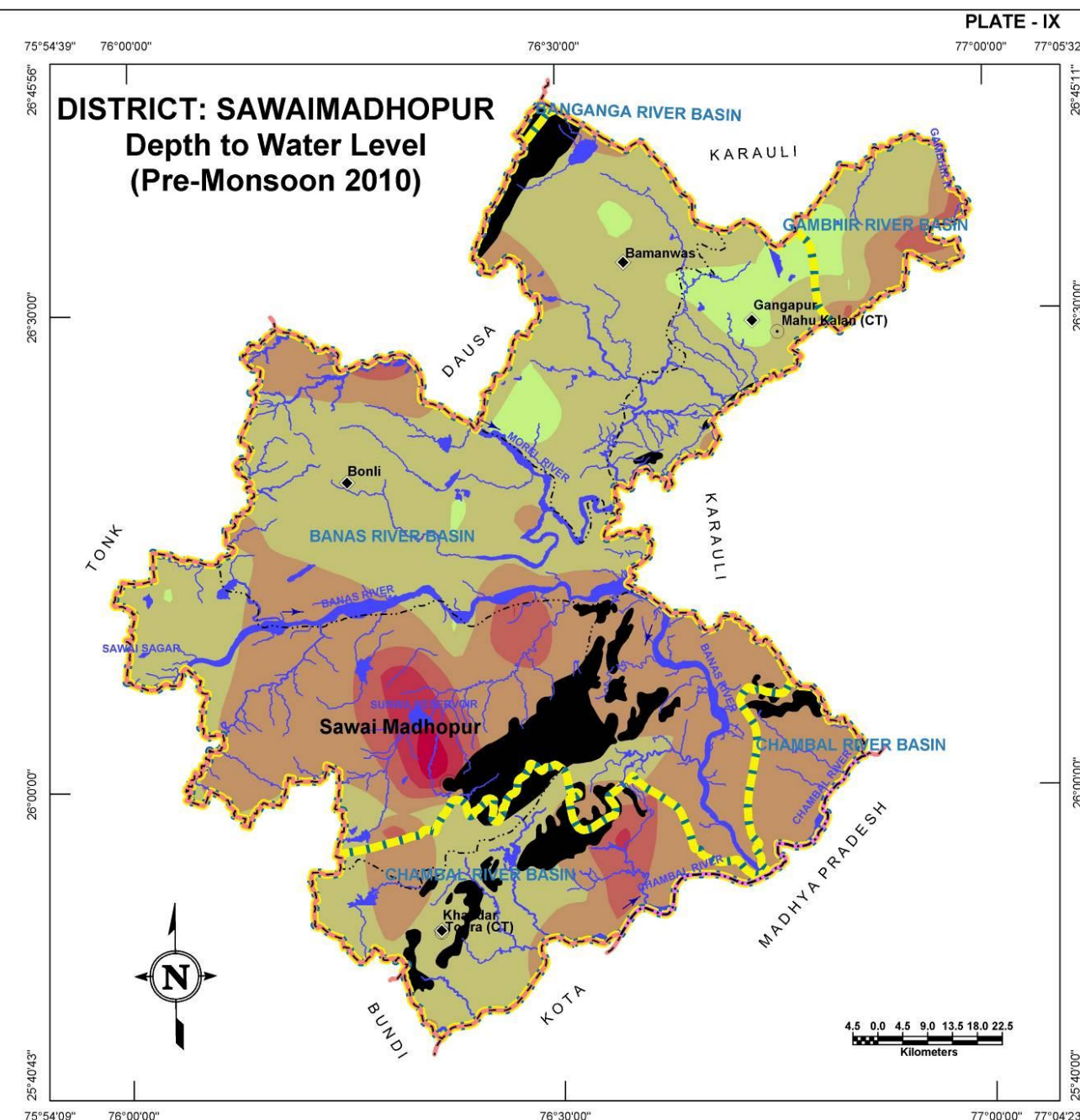
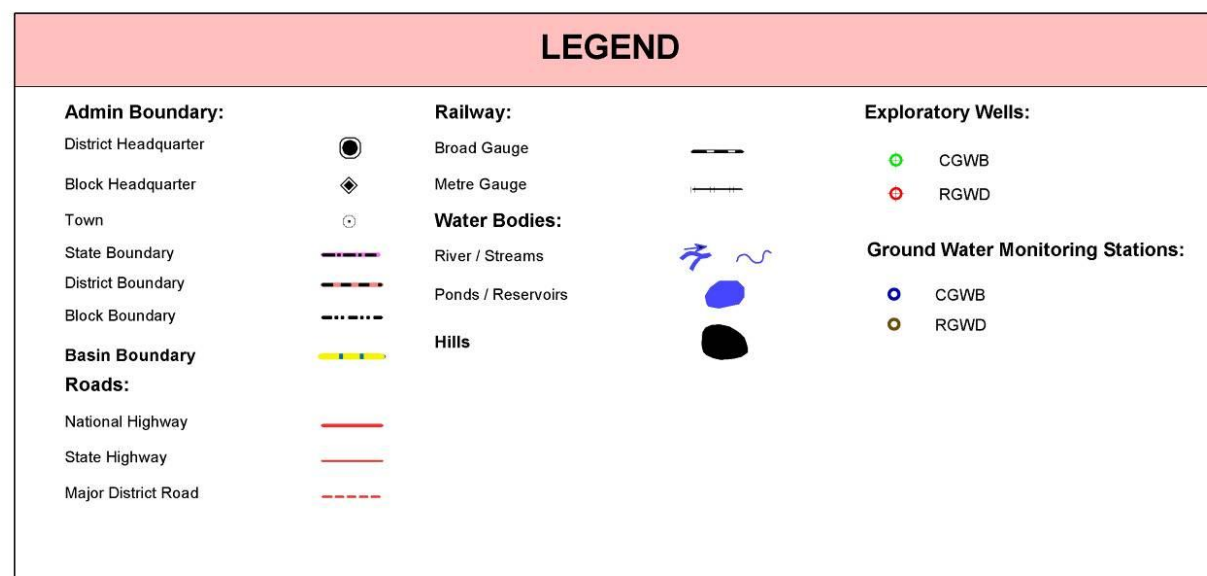
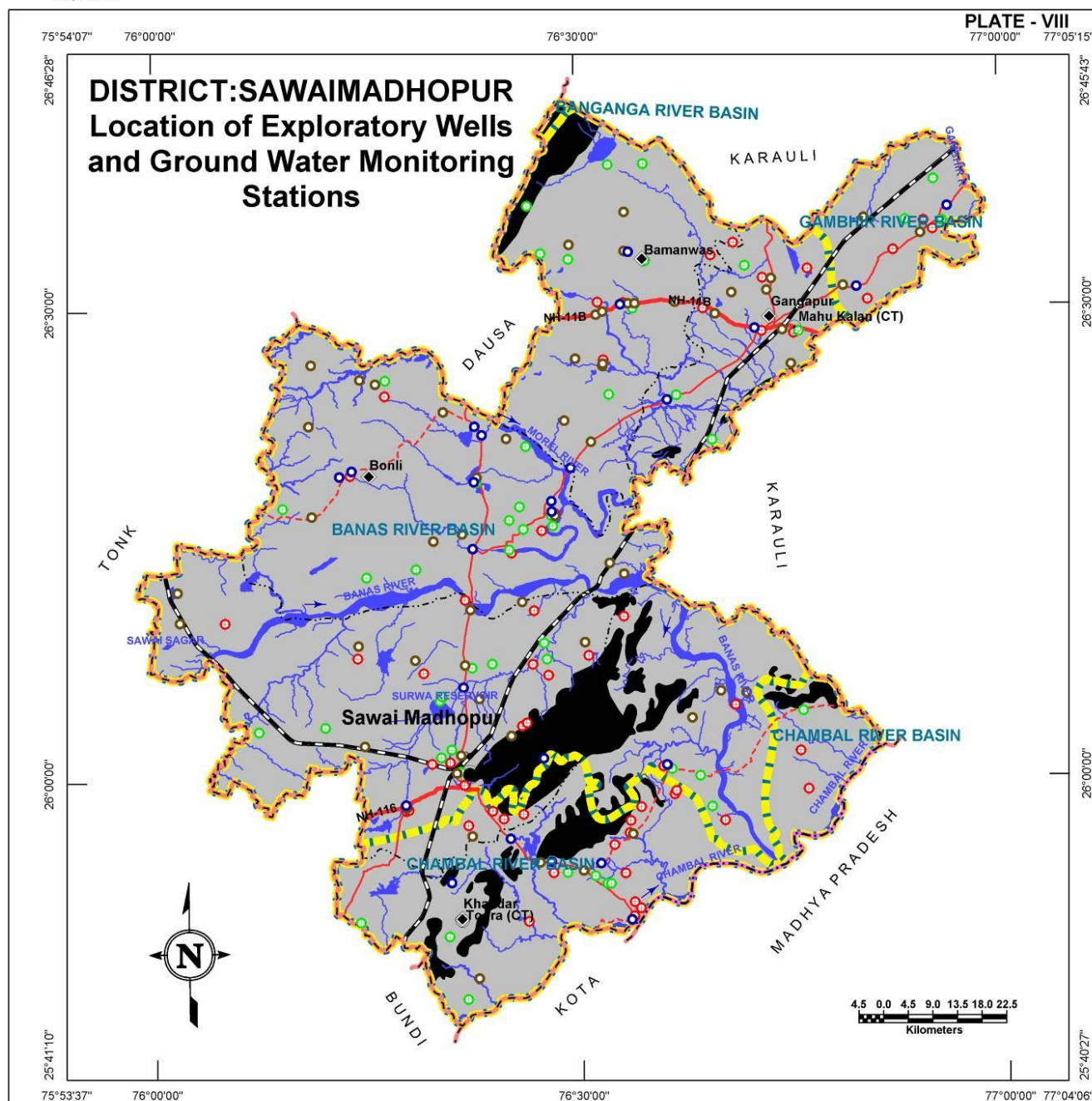
| Block Name     | Exploratory Wells |           |            | Ground Water Monitoring Stations |           |           | Recommended additional wells for optimization of monitoring network |               |
|----------------|-------------------|-----------|------------|----------------------------------|-----------|-----------|---------------------------------------------------------------------|---------------|
|                | CGWB              | RGWD      | Total      | CGWB                             | RGWD      | Total     | Water Level                                                         | Water Quality |
| Bamanwas       | 8                 | 5         | 13         | 3                                | 13        | 16        | 0                                                                   | 27            |
| Bonli          | 13                | 5         | 18         | 8                                | 15        | 23        | 0                                                                   | 17            |
| Gangapur       | 7                 | 9         | 16         | 4                                | 12        | 16        | 0                                                                   | 24            |
| Khandar        | 11                | 18        | 29         | 6                                | 9         | 15        | 1                                                                   | 28            |
| Sawai Madhopur | 11                | 19        | 30         | 3                                | 17        | 20        | 0                                                                   | 19            |
| <b>Total</b>   | <b>50</b>         | <b>56</b> | <b>106</b> | <b>24</b>                        | <b>66</b> | <b>90</b> | <b>1</b>                                                            | <b>115</b>    |

## DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

10m interval has been adopted to depict the depth to ground water levels in Sawai Madhopur district as shown in Plate – IX. Depth to water level varies from less than 10m below ground level to more than 50mbgl. Southern part of the district i.e., Sawai Madhopur – Khandar region shows deeper water levels of 20m - 50m in general, which even reaches to >50m bgl. Northwards, the water level is moderately deep which is about 20-30m bgl in Bamanwas – Gangapur – Bonli region, and some isolated locations in the northern part of Bamanwas and Gangapur blocks have depth to water level at even less than 10m bgl.

| Depth to water level (mbgl) | Block wise area coverage (sq km) * |                |              |                |                | Total Area (sq km) |
|-----------------------------|------------------------------------|----------------|--------------|----------------|----------------|--------------------|
|                             | Bamanwas                           | Bonli          | Gangapur     | Khandar        | Sawai Madhopur |                    |
| < 10                        | 63.8                               | 5.4            | 137.7        | -              | 1.7            | 208.6              |
| 10-20                       | 600.6                              | 733.4          | 390          | 383.3          | 302            | 2,409.3            |
| 20-30                       | 45.8                               | 285.5          | 74.3         | 746.1          | 556.2          | 1,707.9            |
| 30-40                       | -                                  | 24.7           | 23.7         | 63             | 143.8          | 255.2              |
| 40-50                       | -                                  | -              | 0.2          | 3.6            | 71.5           | 75.3               |
| > 50                        | -                                  | -              | -            | -              | 15.9           | 15.9               |
| <b>Total</b>                | <b>710.2</b>                       | <b>1,049.0</b> | <b>625.9</b> | <b>1,196.0</b> | <b>1,091.1</b> | <b>4,672.2</b>     |

\* The area covered in the derived maps is less than the total district area since the hills have been excluded from interpolation/contouring.





## WATER TABLE ELEVATION (PRE MONSOON – 2010)

## DISTRICT – SAWAI MADHOPUR

The water table elevation map as presented in Plate – X demonstrates the distribution of ground water table and its flow direction. The central part around hills east of Sawai Madhopur city and the area in the northwestern part of the district have very high water table elevations i.e., above 280m amsl reaching upto 320m amsl also. The low elevations i.e., <200m amsl are seen in the southeastern part of the district in Khandar block.

**Table: Block wise area covered in each water table elevation range**

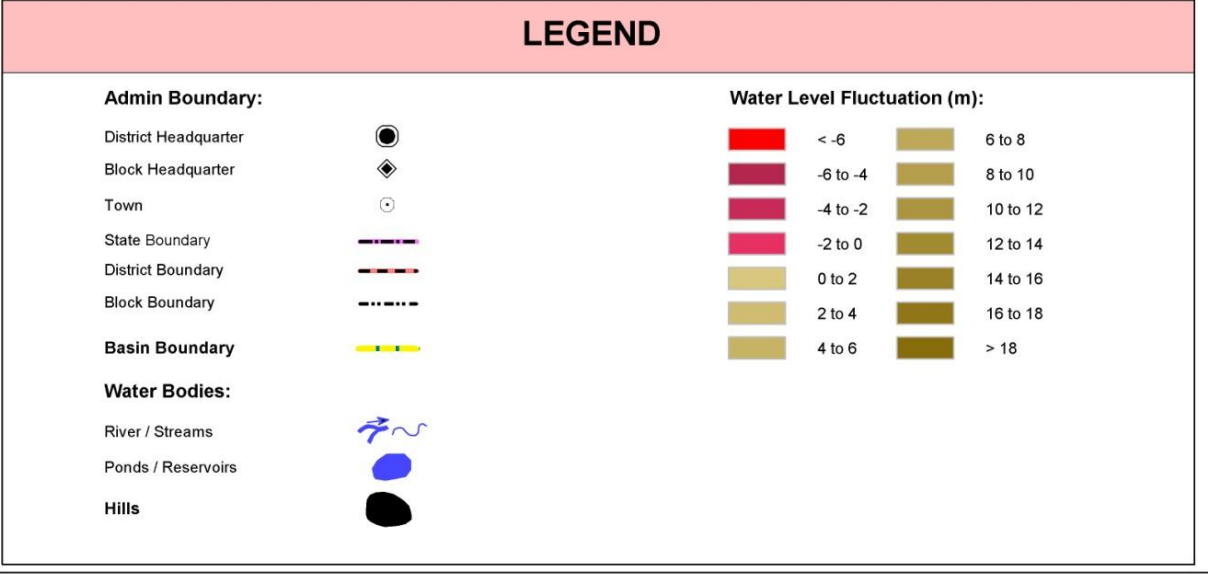
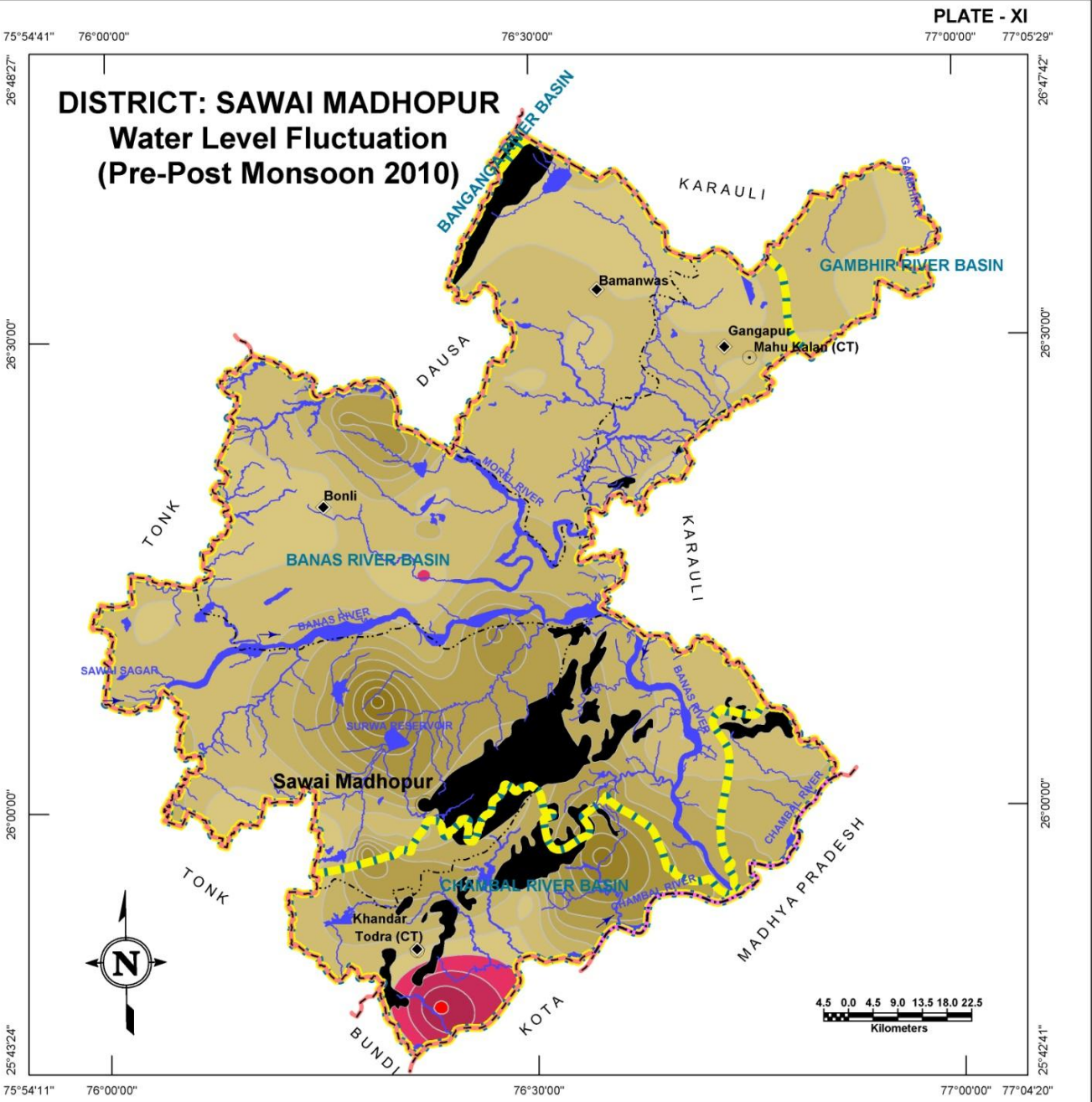
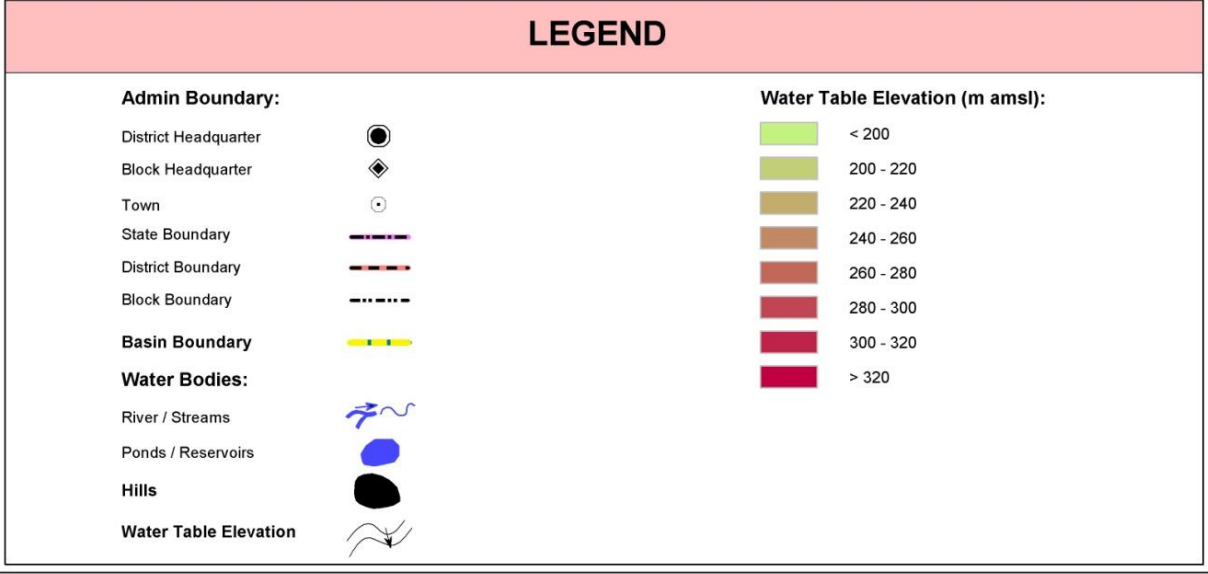
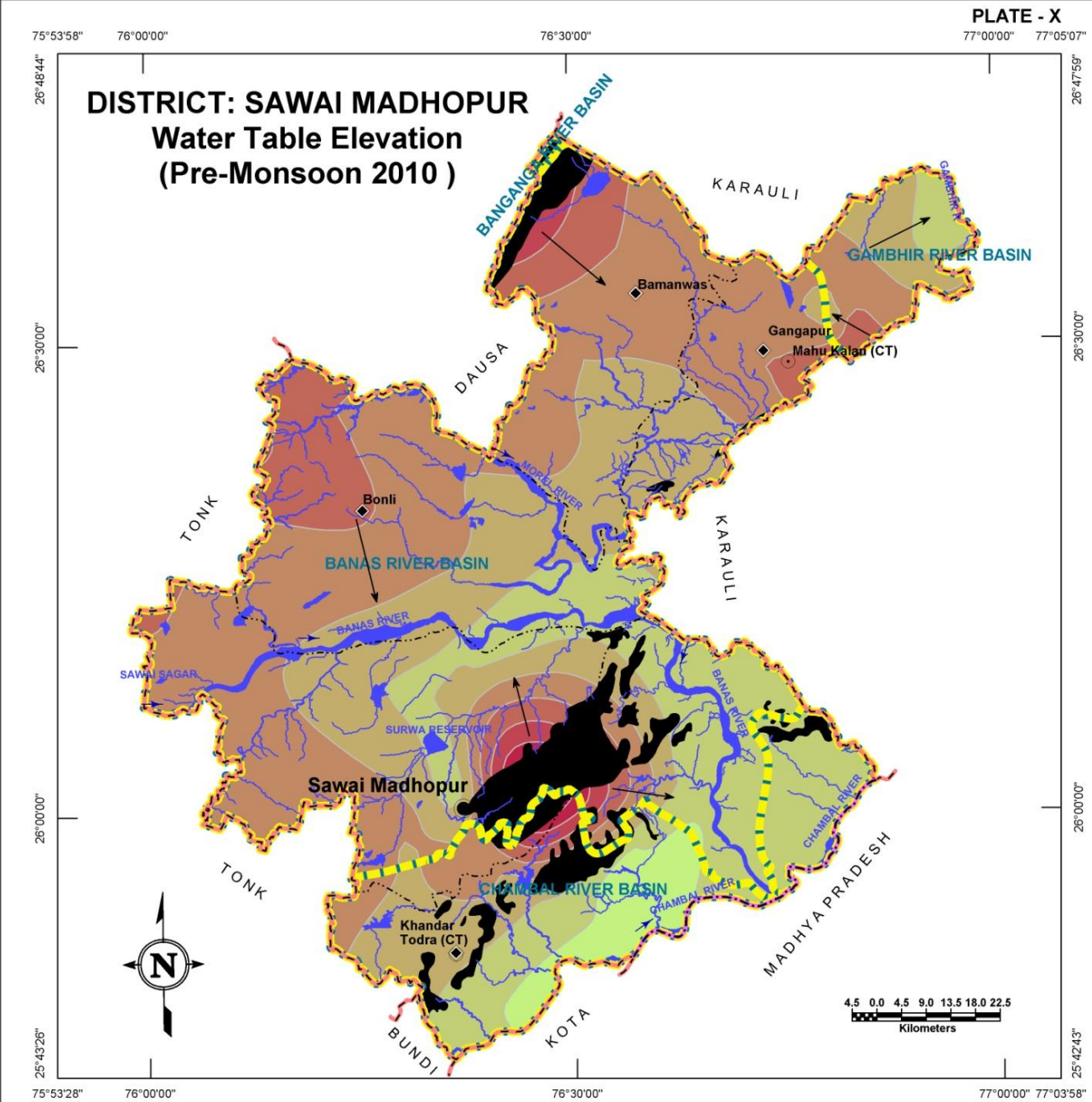
| Water table elevation<br>Range (m amsl) | Block wise area coverage (sq km) |                |              |                |                | Total Area<br>(sq km) |
|-----------------------------------------|----------------------------------|----------------|--------------|----------------|----------------|-----------------------|
|                                         | Bamanwas                         | Bonli          | Gangapur     | Khandar        | Sawai Madhopur |                       |
| < 200                                   | -                                | -              | -            | 183.1          | -              | <b>183.1</b>          |
| 200 - 220                               | 1.7                              | 139.5          | 48.2         | 685.3          | 105.1          | <b>979.9</b>          |
| 220 - 240                               | 191.5                            | 260.0          | 235.4        | 196.9          | 344.1          | <b>1,227.9</b>        |
| 240 - 260                               | 384.6                            | 448.3          | 293.8        | 86.3           | 562.0          | <b>1,775.0</b>        |
| 260 - 280                               | 112.8                            | 198.6          | 48.5         | 22.2           | 51.3           | <b>433.4</b>          |
| 280 - 300                               | 19.6                             | 2.6            | -            | 13.9           | 22.2           | <b>58.3</b>           |
| 300 - 320                               | -                                | -              | -            | 8.1            | 6.1            | <b>14.1</b>           |
| > 320                                   | -                                | -              | -            | 0.2            | 0.3            | <b>0.6</b>            |
| <b>Total</b>                            | <b>710.2</b>                     | <b>1,049.0</b> | <b>625.9</b> | <b>1,196.0</b> | <b>1,091.1</b> | <b>4,672.2</b>        |

## WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

A 2m contour interval reveals ground water level fluctuation in the district from -8 m to +18m as seen in Plate XI. The –ve fluctuation areas (indicated by pink and red regions) are due to over exploitation of ground water in some localized pocket in Khandar block in the southeastern part of the district. Rest of the district has shown a general to significant rise in ground water level in the post monsoon season with respect to pre monsoon season. Maximum rise of more than 18m is noticed in southern part of Sawai Madhopur block. General variation in the district has been in the range of 0m to about 10m in most parts.

**Table: Block wise area covered in each water fluctuation zone**

| Block Name     | Block wise area coverage (sq km) within water level fluctuation range (m) |             |             |             |              |                 |            |              |            |              |             |             |             |            | Total Area<br>(sq km) |
|----------------|---------------------------------------------------------------------------|-------------|-------------|-------------|--------------|-----------------|------------|--------------|------------|--------------|-------------|-------------|-------------|------------|-----------------------|
|                | -8 to -6                                                                  | -6 to -4    | -4 to -2    | -2 to 0     | 0 to 2       | 2 to 4          | 4 to 6     | 6 to 8       | 8 to 10    | 10 to 12     | 12 to 14    | 14 to 16    | 16 to 18    | > 18       |                       |
| Bamanwas       | -                                                                         | -           | -           | -           | 239.8        | 454.1           | 16.3       | -            | -          | -            | -           | -           | -           | -          | <b>710.2</b>          |
| Bonli          | -                                                                         | -           | -           | 1.9         | 225          | 463.7           | 204.3      | 78.4         | 56.8       | 18.9         | -           | -           | -           | -          | <b>1,049.00</b>       |
| Gangapur       | -                                                                         | -           | -           | -           | 239.5        | 386.4           | -          | -            | -          | -            | -           | -           | -           | -          | <b>625.9</b>          |
| Khandar        | 2                                                                         | 24.3        | 40.7        | 41.6        | 47.3         | 323             | 266.7      | 201.4        | 106.7      | 76.1         | 40.2        | 22.2        | 3.8         | -          | <b>1,196.00</b>       |
| Sawai Madhopur | -                                                                         | -           | -           | -           | 66.1         | 336.7           | 160.7      | 133.6        | 217.5      | 101.5        | 42.4        | 21.2        | 9.7         | 1.7        | <b>1,091.10</b>       |
| <b>Total</b>   | <b>2</b>                                                                  | <b>24.3</b> | <b>40.7</b> | <b>43.5</b> | <b>817.7</b> | <b>1,963.90</b> | <b>648</b> | <b>413.4</b> | <b>381</b> | <b>196.5</b> | <b>82.6</b> | <b>43.4</b> | <b>13.5</b> | <b>1.7</b> | <b>4,672.20</b>       |





## GROUND WATER ELECTRICAL CONDUCTIVITY DISTRIBUTION

## DISTRICT – SAWAI MADHOPUR

The electrical conductivity (at 25°C) distribution map is presented in Plate – XII. The areas with low EC values in ground water (<2000  $\mu\text{S}/\text{cm}$ ) are shown in yellow color which occupy almost 83% of the district area indicating that, by and large the ground water in this district is suitable for domestic purposes. The areas with moderately high EC values (2000 -4000  $\mu\text{S}/\text{cm}$ ) are shown in green color and occupy 14% of the district area, largely northern part of the district and around Bonli. Remaining 4% of the district area has high EC values in ground water (>4000  $\mu\text{S}/\text{cm}$ , in red color) largely in the northern part in Gangapur and Bamanwas blocks where the ground water is not suitable for domestic purpose.

**Table: Block wise area of Electrical conductivity distribution**

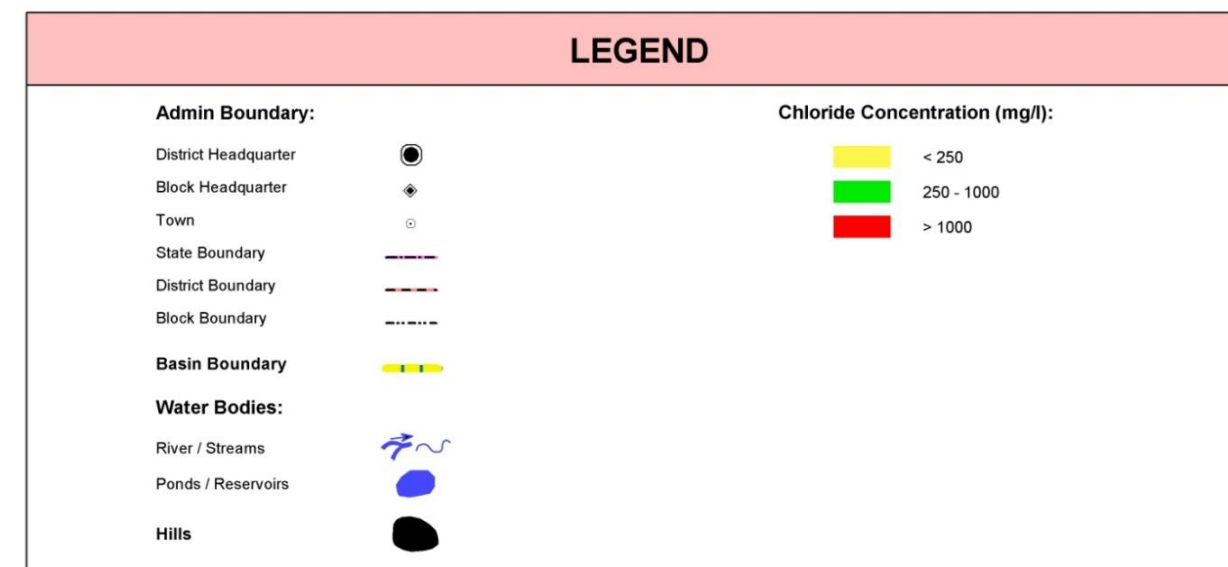
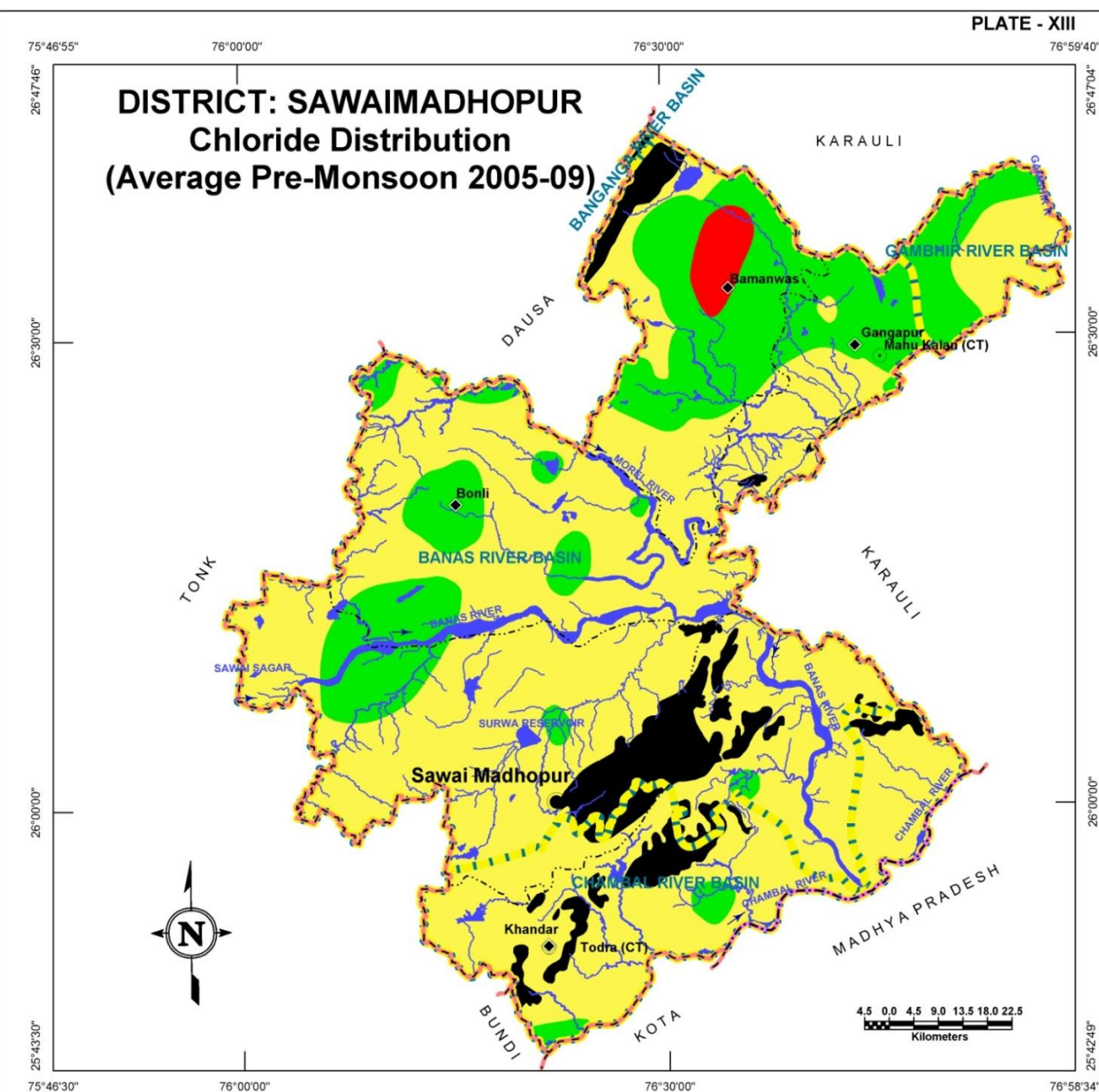
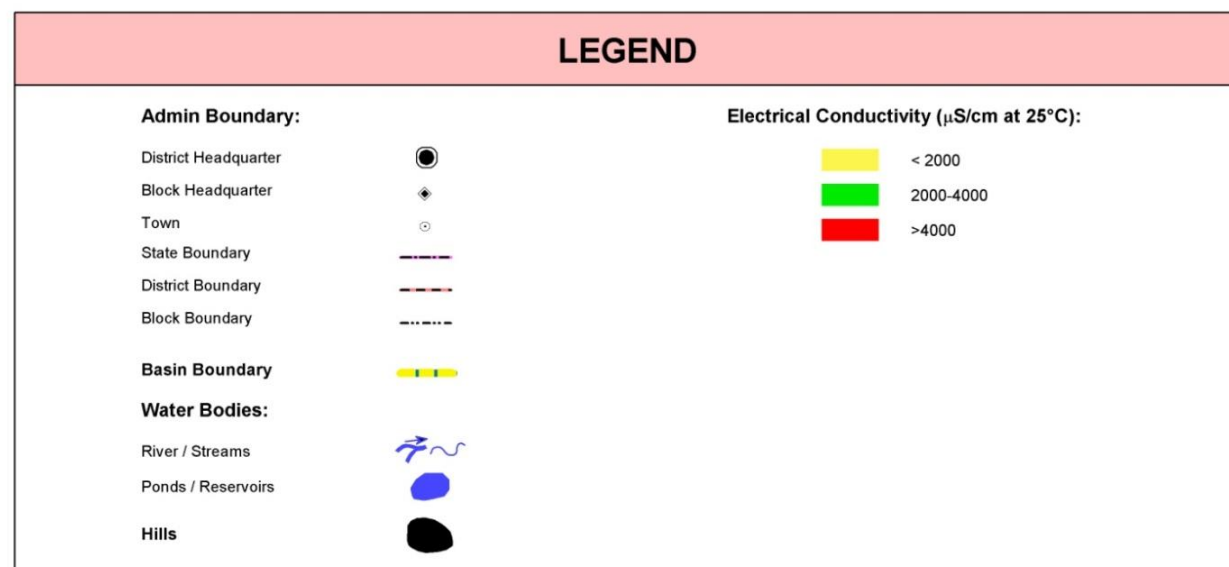
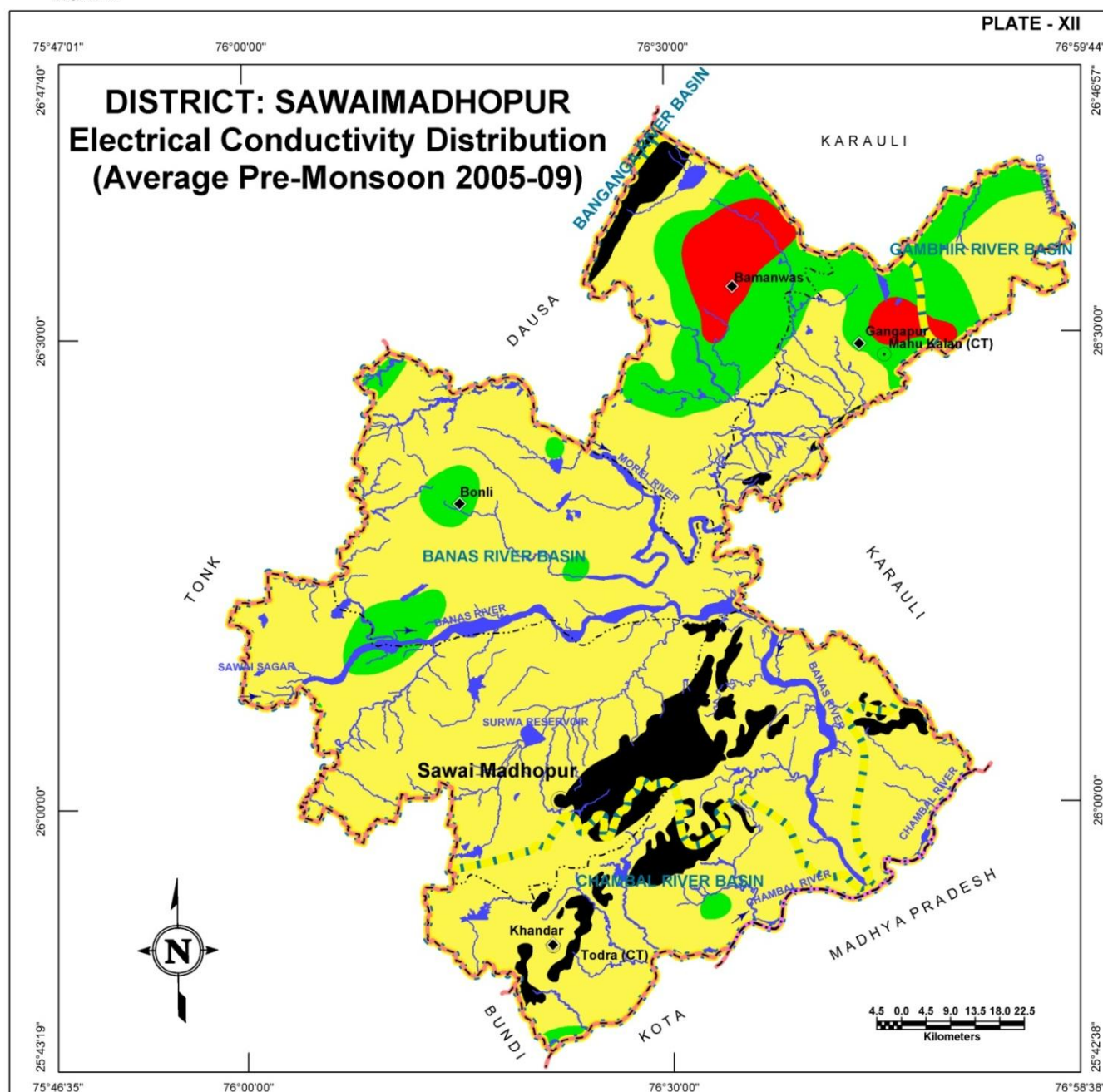
| Electrical Conductivity Ranges<br>(μS/cm at 25°C)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |          |       |         |       |                |       | Total Area<br>(sq km) |
|------------------------------------------------------------------------------|----------------------------------|-------|---------|-------|----------|-------|---------|-------|----------------|-------|-----------------------|
|                                                                              | Bamanwas                         |       | Bonli   |       | Gangapur |       | Khandar |       | Sawai Madhopur |       |                       |
|                                                                              | Area                             | %age  | Area    | %age  | Area     | %age  | Area    | %age  | Area           | %age  |                       |
| < 2000                                                                       | 332.1                            | 46.8  | 934.3   | 89.1  | 362.9    | 58.0  | 1,177.8 | 98.5  | 1,057.4        | 96.9  | 3,864.5               |
| 2000-4000                                                                    | 249.2                            | 35.1  | 114.7   | 10.9  | 224.9    | 35.9  | 18.2    | 1.5   | 33.7           | 3.1   | 640.7                 |
| >4000                                                                        | 128.9                            | 18.1  | -       | -     | 38.1     | 6.1   | -       | -     | -              | -     | 167.0                 |
| Total                                                                        | 710.2                            | 100.0 | 1,049.0 | 100.0 | 625.9    | 100.0 | 1,196.0 | 100.0 | 1,091.1        | 100.0 | 4,672.2               |

## GROUND WATER CHLORIDE DISTRIBUTION

The distribution of chloride concentration is similar in pattern to that of EC. The yellow colored regions in Plate – XIII are low chloride areas (<250 mg/l) occupying approximately 75% of the district area and is suitable for domestic purposes. The areas with moderately high chloride concentration (250-1000mg/l) are shown in green color that occupies approximately 24% of the district area, largely northern part of the district and around Bonli. The remaining negligibly small area (< 0.01%) of the district has high chloride concentration exceeding 1000 mg/l.

**Table: Block wise area of Chloride distribution**

| Chloride Concentration<br>Range (mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |          |       |         |       |                |       | Total Area<br>(sq km) |
|-------------------------------------------------------------------|----------------------------------|-------|---------|-------|----------|-------|---------|-------|----------------|-------|-----------------------|
|                                                                   | Bamanwas                         |       | Bonli   |       | Gangapur |       | Khandar |       | Sawai Madhopur |       |                       |
|                                                                   | Area                             | %age  | Area    | %age  | Area     | %age  | Area    | %age  | Area           | %age  |                       |
| < 250                                                             | 266.6                            | 38.0  | 793.1   | 76.0  | 316.9    | 51.0  | 1,148.9 | 96.0  | 958.4          | 87.9  | 3,483.9               |
| 250-1000                                                          | 378.0                            | 53.0  | 255.9   | 24.0  | 308.9    | 49.0  | 47.1    | 4.0   | 132.7          | 12.1  | 1,122.6               |
| > 1000                                                            | 65.6                             | 9.0   | -       | -     | 0.1      | -     | -       | -     | -              | -     | 65.7                  |
| Total                                                             | 710.2                            | 100.0 | 1,049.0 | 100.0 | 625.9    | 100.0 | 1,196.0 | 100.0 | 1,091.1        | 100.0 | 4,672.2               |





## GROUND WATER FLUORIDE DISTRIBUTION

## DISTRICT – SAWAI MADHOPUR

The Fluoride concentration map is presented in Plate – XIV. The areas with low concentration (i.e., >1.5 mg/l) are shown in yellow color which occupy almost 76% of the district area which is suitable for domestic purpose. The areas with moderately high concentration (1.5-3.0 mg/l) are shown in green color that occupies approximately 19% of the district area, largely in the northern and western part of the district. Remaining part of the district (approximately 5%) has high Fluoride concentration (>3.0 mg/l) as shown in red color. These high Fluoride areas are scattered as four different patches in western and northwestern parts of the district and unsuitable for domestic purposes.

**Table: Block wise area of Fluoride distribution**

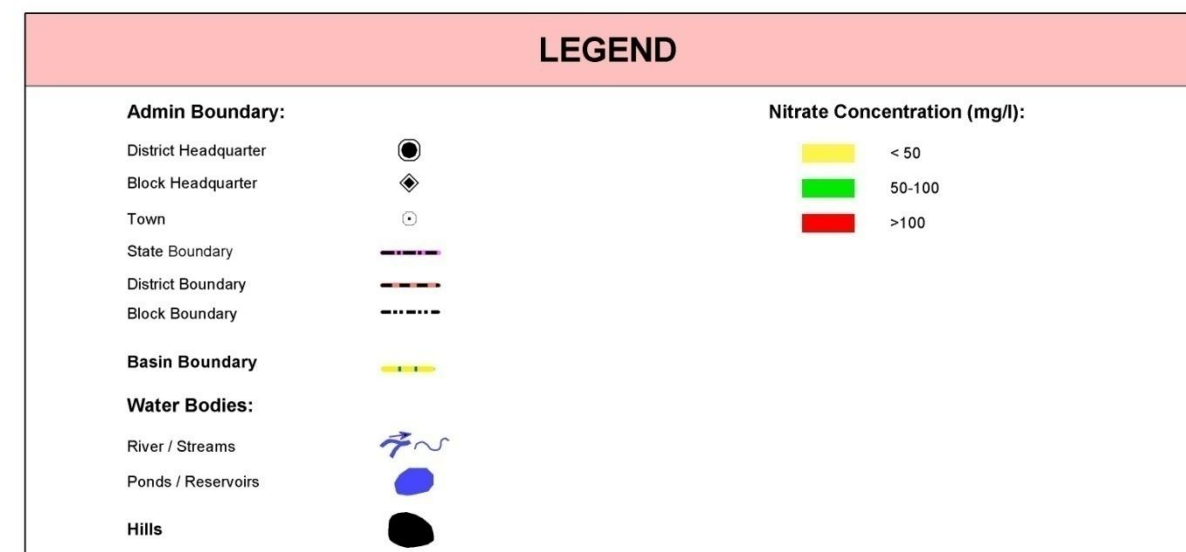
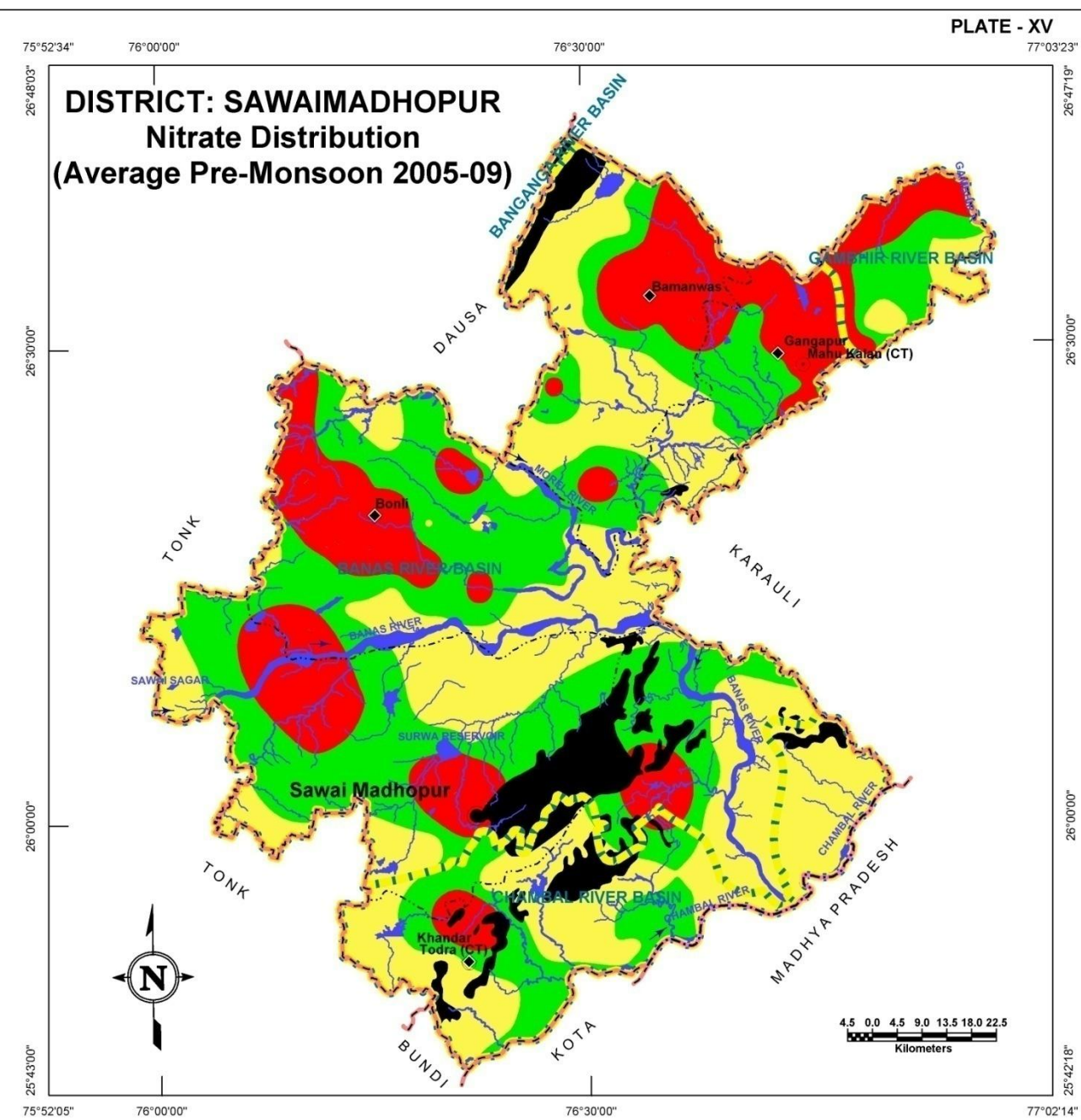
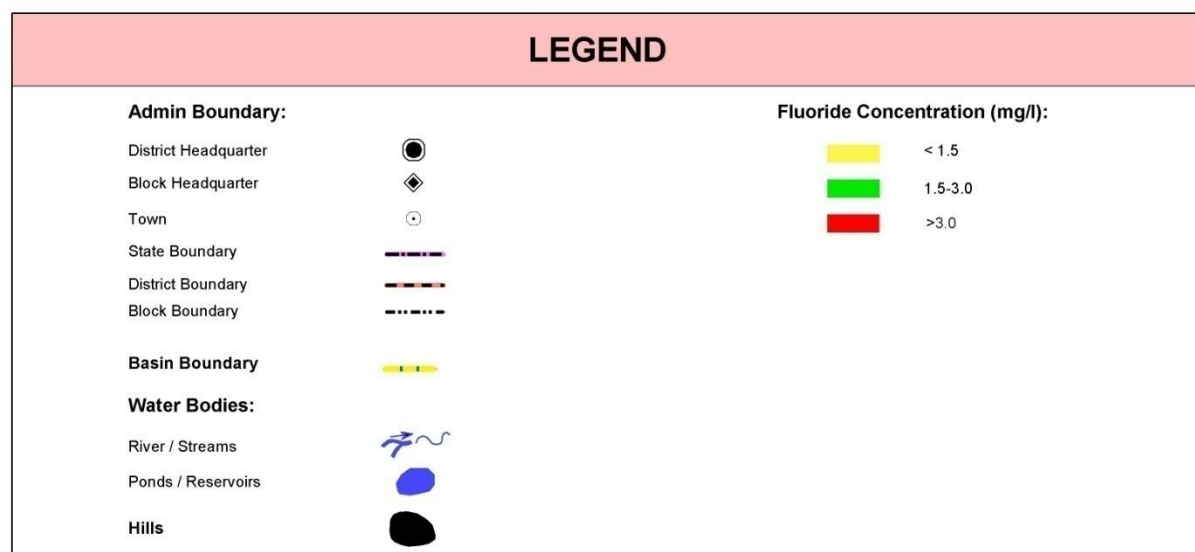
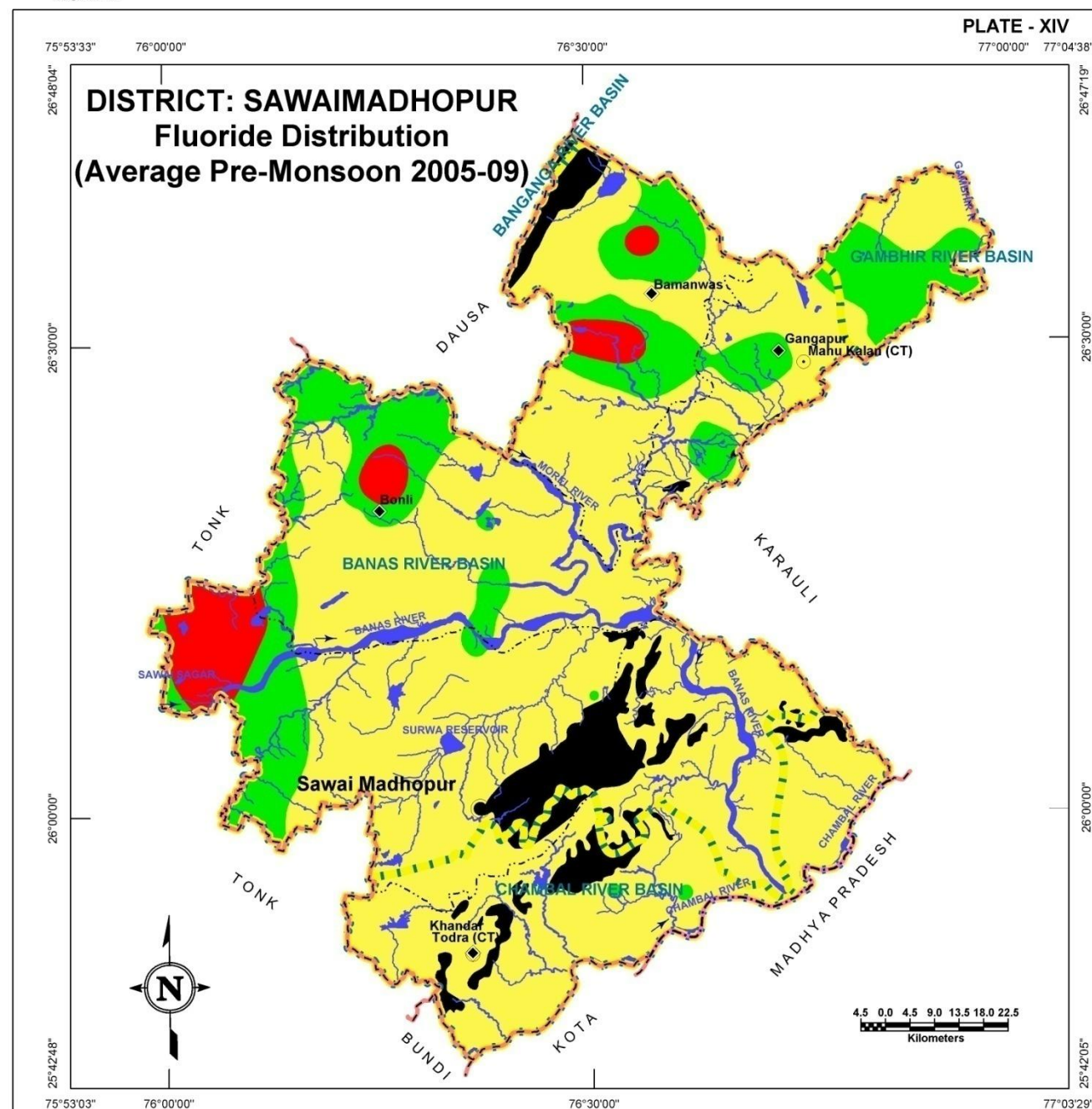
| Fluoride concentration<br>Range (mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |          |       |         |       |                |       | Total Area<br>(sq km) |
|-------------------------------------------------------------------|----------------------------------|-------|---------|-------|----------|-------|---------|-------|----------------|-------|-----------------------|
|                                                                   | Bamanwas                         |       | Bonli   |       | Gangapur |       | Khandar |       | Sawai Madhopur |       |                       |
|                                                                   | Area                             | %age  | Area    | %age  | Area     | %age  | Area    | %age  | Area           | %age  |                       |
| < 1.5                                                             | 454.5                            | 64.0  | 739.4   | 70.5  | 392.6    | 62.7  | 1,190.8 | 99.6  | 775.8          | 71.1  | 3,553.1               |
| 1.5-3.0                                                           | 204.6                            | 28.8  | 272.5   | 26.0  | 233.3    | 37.3  | 5.2     | 0.4   | 185.6          | 17.0  | 901.2                 |
| > 3.0                                                             | 51.1                             | 7.2   | 37.1    | 3.5   | -        | -     | -       | -     | 129.7          | 11.9  | 217.9                 |
| Total                                                             | 710.2                            | 100.0 | 1,049.0 | 100.0 | 625.9    | 100.0 | 1,196.0 | 100.0 | 1,091.1        | 100.0 | 4,672.2               |

## GROUND WATER NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate–XV shows distribution of Nitrate in ground water. High nitrate concentration (>100 mg/l) areas are seen in red colour in the map which occupies about 23% of the district area in scattered patches spread all over the district resulting into the ground water in such areas remaining unsuitable for agriculture purpose. Moderately high nitrate concentration (50 – 100 mg/l) is shown in green color (38% of the district area) and the areas with low concentration nitrate (<50 mg/l) are shown in yellow color seen to occupy approximately 39% of the district area and the latter being most suitable for agriculture.

**Table: Block wise area of Nitrate distribution**

| Nitrate concentration<br>Range(mg/l)<br>(Ave. of years 2005-09) | Block wise area coverage (sq km) |       |         |       |          |       |         |       |                |       | Total Area<br>(sq km) |
|-----------------------------------------------------------------|----------------------------------|-------|---------|-------|----------|-------|---------|-------|----------------|-------|-----------------------|
|                                                                 | Bamanwas                         |       | Bonli   |       | Gangapur |       | Khandar |       | Sawai Madhopur |       |                       |
|                                                                 | Area                             | %age  | Area    | %age  | Area     | %age  | Area    | %age  | Area           | %age  |                       |
| < 50                                                            | 319.2                            | 45.0  | 283.2   | 27.0  | 140.5    | 22.5  | 732.8   | 61.3  | 362.2          | 33.2  | 1,837.9               |
| 50-100                                                          | 185.5                            | 26.1  | 463.9   | 44.2  | 230.3    | 36.7  | 381.1   | 31.8  | 507.8          | 46.5  | 1,768.6               |
| >100                                                            | 205.5                            | 28.9  | 301.9   | 28.8  | 255.1    | 40.8  | 82.1    | 6.9   | 221.1          | 20.3  | 1,065.7               |
| Total                                                           | 710.2                            | 100.0 | 1,049.0 | 100.0 | 625.9    | 100.0 | 1,196.0 | 100.0 | 1,091.1        | 100.0 | 4,672.2               |





## DEPTH TO BEDROCK

Plate–XVI depicts the distribution of bedrock depth below ground level. The thick alluvial deposits are underlain by bedrock of different lithology and age. The beginning of massive bedrock has been considered for defining top of bedrock surface. The major rocks types occurring in the district are Phyllite, Limestone, Shale and Quartzite. These rocks are overlain by alluvial deposits of sand, clay, silt and admixture of these in different proportions and thicknesses. The map reveals that the bedrock surface is highly undulating in the northeastern part whereas is relatively flat towards west and southwest varying from about 40m bgl to more than 60m bgl. Shallow bedrock depth is found in the Sawai Madhopur area (less than 20m bgl). Deepest occurrence of bedrock (indicating high alluvial thickness) is found in Gangapur and Khandar blocks where the depth of occurrence observed more than 60m bgl.

| Depth to bedrock<br>(m bgl) | Block wise area coverage (sq km) |       |        |       |          |       |         |       |                |       | Total Area<br>(sq km) |
|-----------------------------|----------------------------------|-------|--------|-------|----------|-------|---------|-------|----------------|-------|-----------------------|
|                             | Bamanwas                         |       | Bonli  |       | Gangapur |       | Khandar |       | Sawai Madhopur |       |                       |
|                             | Area                             | %age  | Area   | %age  | Area     | %age  | Area    | %age  | Area           | %age  |                       |
| < 20                        | 0                                | 0     | 0      | 0     | 0        | 0     | 0       | 0     | 2.6            | 0     | 2.6                   |
| 20-40                       | 465.6                            | 65.6  | 972.8  | 93.0  | 58.1     | 9.3   | 1,014.6 | 84.8  | 1009.2         | 93.0  | 3,520.3               |
| 40-60                       | 244.6                            | 34.4  | 76.2   | 7.0   | 520.9    | 83.3  | 181.4   | 15.2  | 79.3           | 7.0   | 1,102.4               |
| > 60                        | 0                                | 0     | 0      | 0     | 46.9     | 7.4   | 0       | 0     | 0              | 0     | 46.9                  |
| Total                       | 710.2                            | 100.0 | 1049.0 | 100.0 | 625.9    | 100.0 | 1196.0  | 100.0 | 1091.1         | 100.0 | 4,672.2               |

## UNCONFINED AQUIFER

### Alluvial areas

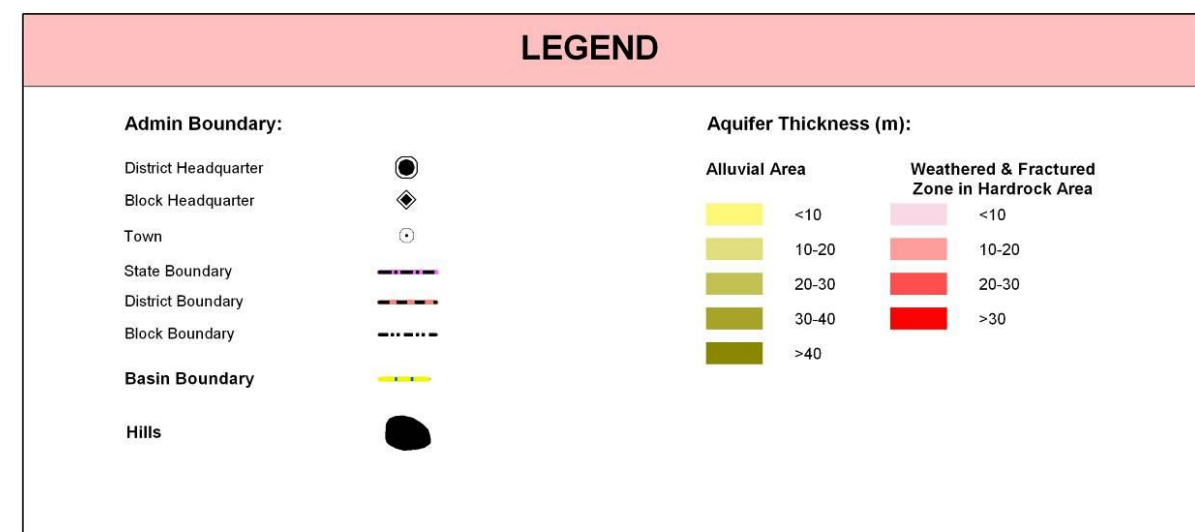
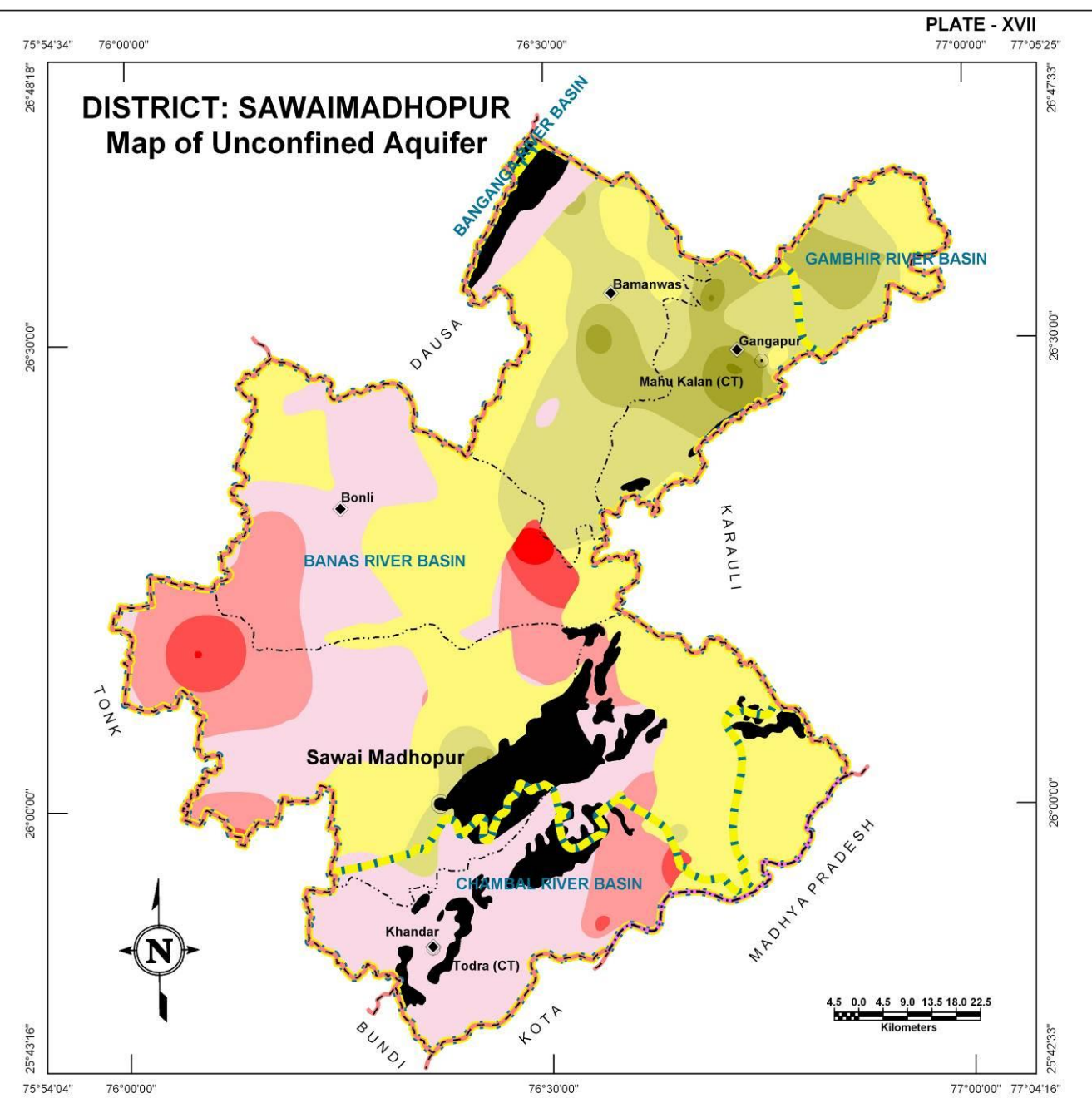
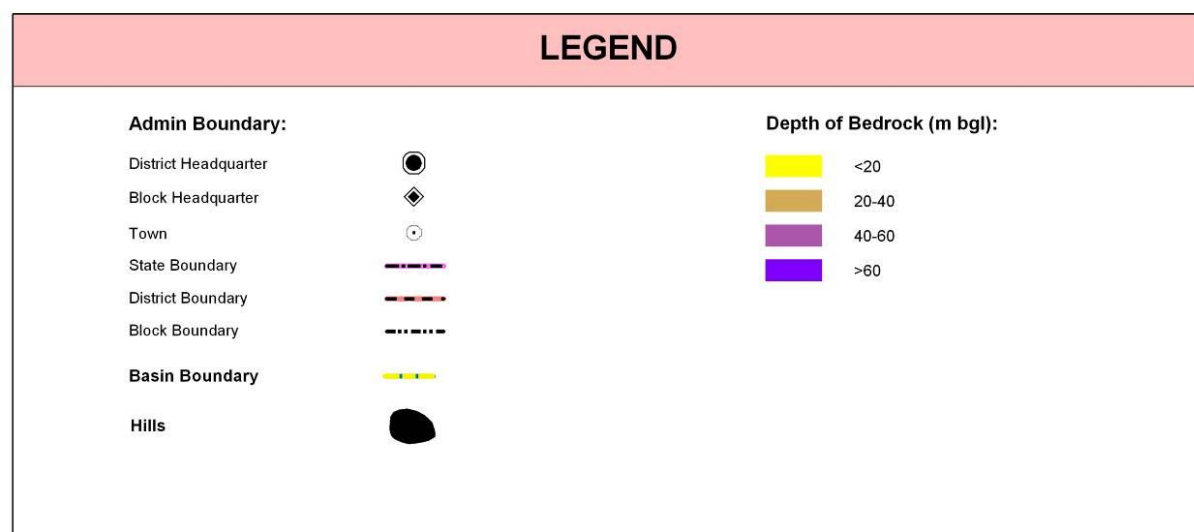
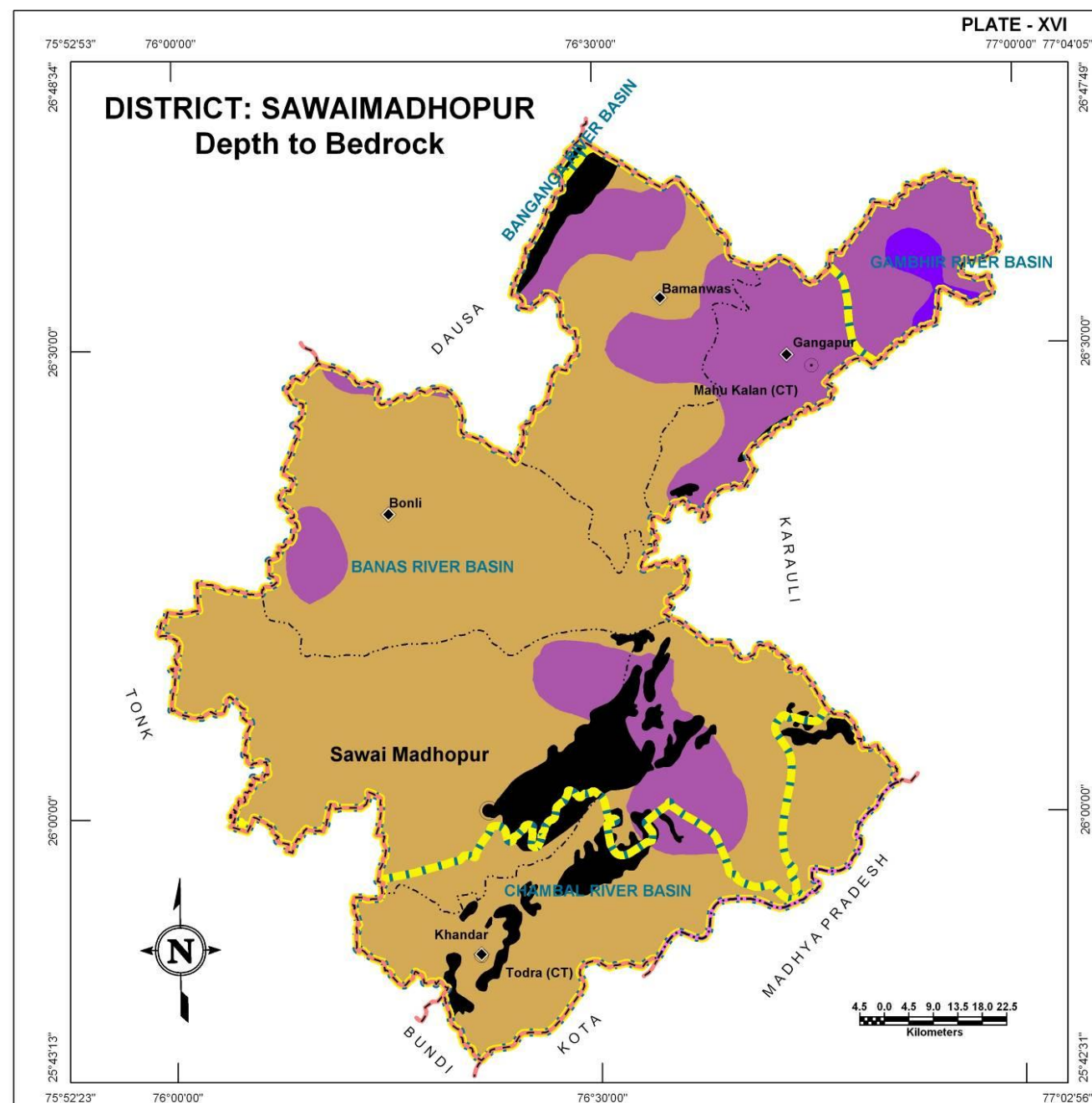
Northernmost part of the district has thick alluvial aquifers forming unconfined aquifers there with 20m to more than 40m thickness. Perusal of Plate – XVII reveals a moderate thickness of upto 30m bgl in the area with pockets of more thick (>40m) unconfined alluvial aquifer material in the MahuKalan city. Shallow thicknesses of alluvial materials are encountered towards southeastern, central and western fringe of the district with less than 10m thickness.

| Unconfined aquifer<br>Thickness (m) | Block wise Area coverage (sq km) |              |              |              |                | Total Area<br>(sq km) |
|-------------------------------------|----------------------------------|--------------|--------------|--------------|----------------|-----------------------|
|                                     | Bamanwas                         | Bonli        | Gangapur     | Khandar      | Sawai Madhopur |                       |
| < 10                                | 203.7                            | 457.7        | 96.0         | 506.4        | 303.8          | 1,567.6               |
| 10-20                               | 339.3                            | 26.7         | 283.0        | 4.0          | 58.5           | 711.5                 |
| 20-30                               | 89.2                             | -            | 192.8        | -            | 7.4            | 289.4                 |
| 30-40                               | 8.5                              | -            | 52.0         | -            | 0.2            | 60.7                  |
| > 40                                | -                                | -            | 2.1          | -            | -              | 2.1                   |
| <b>Total</b>                        | <b>640.7</b>                     | <b>484.4</b> | <b>625.9</b> | <b>510.4</b> | <b>369.9</b>   | <b>2,631.3</b>        |

### Hard rock areas

Weathered, fractured and jointed rock formations occurring at shallower depths constitute good unconfined aquifers. Such aquifers are mainly located in the western, central and southern parts of the district with isolated occurrence in the northwestern part ranging in thickness from less than 10 meter to more than 30 meter. The maximum thickness of aquifer materials in hardrock (more than 30m) has been reported from Bamanwas, Bonli and Sawai Madhopur blocks however, covering very limited area.

| Unconfined aquifer<br>Thickness (m) | Block wise Area coverage (sq km) |              |          |              |                | Total Area<br>(sq km) |
|-------------------------------------|----------------------------------|--------------|----------|--------------|----------------|-----------------------|
|                                     | Bamanwas                         | Bonli        | Gangapur | Khandar      | Sawai Madhopur |                       |
| < 10                                | 68.2                             | 355.4        | -        | 525.0        | 371.9          | 1,320.5               |
| 10-20                               | -                                | 157.6        | -        | 147.8        | 276.6          | 582.0                 |
| 20-30                               | 0.8                              | 35.8         | -        | 12.8         | 72.1           | 121.5                 |
| > 30                                | 0.5                              | 15.8         | -        | -            | 0.6            | 16.9                  |
| <b>Total</b>                        | <b>69.5</b>                      | <b>564.6</b> | <b>-</b> | <b>685.6</b> | <b>721.2</b>   | <b>2,040.9</b>        |





## Glossary of terms

| S. No. | Technical Terms        | Definition                                                                                                                                   |
|--------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | AQUIFER                | A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.          |
| 2      | ARID CLIMATE           | Climate characterized by high evaporation and low precipitation.                                                                             |
| 3      | ARTIFICIAL RECHARGE    | Addition of water to a groundwater reservoir by man-made activity                                                                            |
| 4      | CLIMATE                | The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region. |
| 5      | CONFINED AQUIFER       | A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.                 |
| 6      | CONTAMINATION          | Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.                      |
| 7      | DRAWDOWN               | The drawdown is the depth by which water level is lowered.                                                                                   |
| 8      | FRESH WATER            | Water suitable for drinking purpose.                                                                                                         |
| 9      | GROUND WATER           | Water found below the land surface.                                                                                                          |
| 10     | GROUND WATER BASIN     | A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.                                           |
| 11     | GROUNDWATER RECHARGE   | The natural infiltration of surface water into the ground.                                                                                   |
| 12     | HARD WATER             | The water which does not produce sufficient foam with soap.                                                                                  |
| 13     | HYDRAULIC CONDUCTIVITY | A constant that serves as a measure of permeability of porous medium.                                                                        |
| 14     | HYDROGEOLOGY           | The science related with the ground water.                                                                                                   |
| 15     | HUMID CLIMATE          | The area having high moisture content.                                                                                                       |
| 16     | ISOHYET                | A line of equal amount of rainfall.                                                                                                          |
| 17     | METEOROLOGY            | Science of the atmosphere.                                                                                                                   |
| 18     | PERCOLATION            | It is flow through a porous substance.                                                                                                       |
| 19     | PERMEABILITY           | The property or capacity of a soil or rock for transmitting water.                                                                           |
| 20     | pH                     | Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).                               |
| 21     | PIEZOMETRIC HEAD       | Elevation to which water will rise in a piezometers.                                                                                         |
| 22     | RECHARGE               | It is a natural or artificial process by which water is added from outside to the aquifer.                                                   |
| 23     | SAFE YIELD             | Amount of water which can be extracted from groundwater without producing undesirable effect.                                                |
| 24     | SALINITY               | Concentration of dissolved salts.                                                                                                            |
| 25     | SEMI-ARID              | An area is considered semiarid having annual rainfall between 10-20 inches.                                                                  |
| 26     | SEMI-CONFINED AQUIFER  | Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.                                                                  |
| 27     | SPECIFIC YIELD         | Quantity of water which is released by a formation after its complete saturation.                                                            |
| 28     | TOTAL DISSOLVED SOLIDS | Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.                                  |

(Contd...)

| S. No. | Technical Terms            | Definition                                                                                                                                                                                                                   |
|--------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29     | TRANSMISSIBILITY           | It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability. |
| 30     | UNCONFINED AQUIFER         | A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.                                                                                                              |
| 31     | UNSATURATED ZONE           | The zone below the land surface in which pore space contains both water and air.                                                                                                                                             |
| 32     | WATER CONSERVATION         | Optimal use and proper storage of water.                                                                                                                                                                                     |
| 33     | WATER RESOURCES            | Availability of surface and ground water.                                                                                                                                                                                    |
| 34     | WATER RESOURCES MANAGEMENT | Planned development, distribution and use of water resources.                                                                                                                                                                |
| 35     | WATER TABLE                | Water table is the upper surface of the zone of saturation at atmospheric pressure.                                                                                                                                          |
| 36     | ZONE OF SATURATION         | The ground in which all pores are completely filled with water.                                                                                                                                                              |
| 37     | ELECTRICAL CONDUCTIVITY    | Flow of free ions in the water at 25C mu/cm.                                                                                                                                                                                 |
| 38     | CROSS SECTION              | A Vertical Projection showing sub-surface formations encountered in a specific plane.                                                                                                                                        |
| 39     | 3-D PICTURE                | A structure showing all three dimensions i.e. length, width and depth.                                                                                                                                                       |
| 40     | GWD                        | Ground Water Department                                                                                                                                                                                                      |
| 41     | CGWB                       | Central Ground Water Board                                                                                                                                                                                                   |
| 42     | CGWA                       | Central Ground Water Authority                                                                                                                                                                                               |
| 43     | SWRPD                      | State Water Resources Planning Department                                                                                                                                                                                    |
| 44     | EU-SPP                     | European Union State Partnership Programme                                                                                                                                                                                   |
| 45     | TOPOGRAPHY                 | Details of drainage lines and physical features of land surface on a map.                                                                                                                                                    |
| 46     | GEOLOGY                    | The science related with the Earth.                                                                                                                                                                                          |
| 47     | GEOMORPHOLOGY              | The description and interpretation of land forms.                                                                                                                                                                            |
| 48     | PRE MONSOON SURVEY         | Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)                                                                                                 |
| 49     | POST-MONSOON SURVEY        | Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)                                                                                          |
| 50     | PIEZOMETER                 | A non-pumping small diameter bore hole used for monitoring of static water level.                                                                                                                                            |
| 51     | GROUND WATER FLUCTUATION   | Change in static water level below ground level.                                                                                                                                                                             |
| 52     | WATER TABLE                | The static water level found in unconfined aquifer.                                                                                                                                                                          |
| 53     | DEPTH OF BED ROCK          | Hard & compact rock encountered below land Surface.                                                                                                                                                                          |
| 54     | G.W. MONITORING STATION    | Dug wells selected on grid basis for monitoring of state water level.                                                                                                                                                        |
| 55     | EOLIAN DEPOSITS            | Wind-blown sand deposits                                                                                                                                                                                                     |

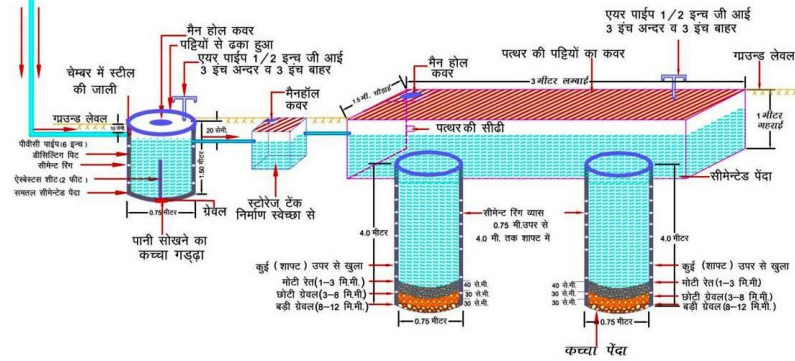




भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक  
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

चित्र-4

- PVC पाईप 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.50 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 3 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्भरित जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 40,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



## भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरॉन-स्नायु तंत्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त परिसंचरण रोगों से ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

नाइट्रेट-नवजात शिशुओं में ब्लू बेबी बीमारी (मेथेमोग्लोबिनिमिया)

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक व मानसिक विकास में बाधा वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न, हड्डियों में मुड़ाव



केन्द्रीय भूमि जल बोर्ड,  
पश्चिमी क्षेत्र, जयपुर  
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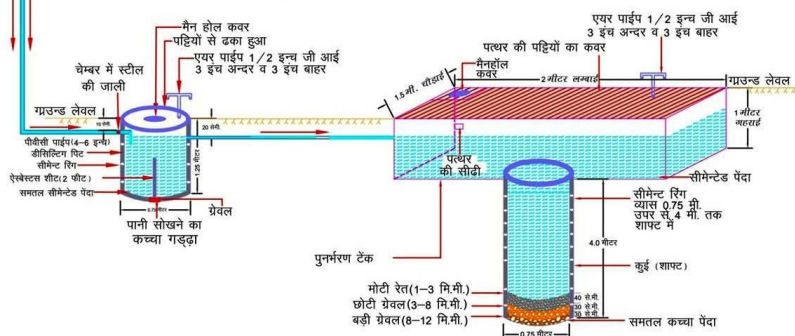
भूजल अमूल्य है इसे प्रदूषित न करें।



भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक  
निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

चित्र-3

- PVC पाईप 4" - 6" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- रीचार्ज टैंक 1.5 मी चौड़ा x 2 मी लम्बा x 1 मी गहरा
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 15,000-16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

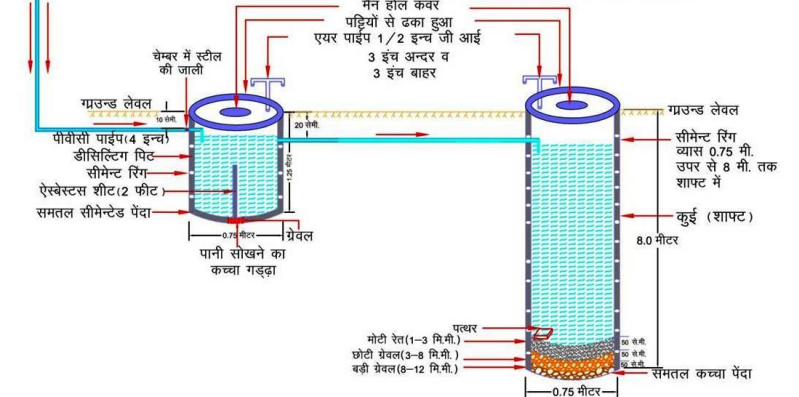


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 11,000-12,000
- वार्षिक पुनर्भरित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 16,64,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

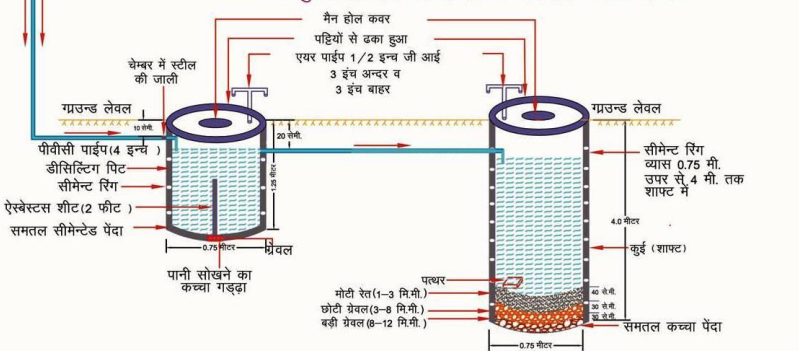


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

- PVC पाईप 4" व्यास
- सीमेंट रिंग से निर्मित डीसिल्टिंग पिट (0.75 मी व्यास x 1.25 मी गहरा)
- सीमेंट रिंग से निर्मित शाफ्ट (0.75 मी व्यास x 4 मी गहरा)
- संरचना की अनुमानित लागत रु 7000-8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम





## Myths and Facts about Ground Water

| S No | Myths                                                                                                                                                                        | Facts                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | What is Ground Water <ul style="list-style-type: none"> <li>• an underground lake</li> <li>• a net work of underground rivers</li> <li>• a bowl filled with water</li> </ul> | Water which occurs below the land in geological formations/rocks is Ground water                                                                                                       |
| 2    | Ground Water occurs everywhere beneath the Land Surface                                                                                                                      | Not really, it depends on the nature of rock formation                                                                                                                                 |
| 3    | There is a relationship between ground water and surface water                                                                                                               | Not all the places. Near streams/rivers there is relation                                                                                                                              |
| 4    | Groundwater is not renewable resource                                                                                                                                        | It is renewable source and every year it is being recharged through rain/applied irrigation etc                                                                                        |
| 5    | Ground water is unlimited and deeper you drill more discharge                                                                                                                | It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper                                                                         |
| 6    | Ground Water moves rapidly                                                                                                                                                   | The movement of ground water is very slow                                                                                                                                              |
| 7    | Ground water pumped from wells is thousands of years old                                                                                                                     | Generally the ground water being tapped through wells is a few years old                                                                                                               |
| 8    | If water taste good—it is safe to drink                                                                                                                                      | It may have other chemicals e.g. fluoride, nitrates etc which are harmful                                                                                                              |
| 9    | Water from free flowing tube wells is very pure                                                                                                                              | This water can also be contaminated so test before use                                                                                                                                 |
| 10   | If I recharge my TW/DW/HP it will not benefit me                                                                                                                             | It will also benefit you and also adjoining wells                                                                                                                                      |
| 11   | There is no static ground water resources in Rajasthan                                                                                                                       | Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge                                                   |
| 12   | I cannot meet annual cooking and drinking water requirement by rain water harvesting                                                                                         | The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement |
| 13   | You can increase ground water recharge                                                                                                                                       | This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures                                                        |
| 14   | You cannot use abandoned TW/HP/DW for ground water recharge                                                                                                                  | These should be used as recharge structures as harvested rain water is directly put into GW reservoir                                                                                  |
| 15   | Putting waste near HP/TW will not cause any problem                                                                                                                          | Such actions will pollute wells and water                                                                                                                                              |



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